Evaluating the Development Impacts of Climate Finance

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Abstract

While foreign aid explicitly targets poverty-related development outcomes, climate finance does not. This raises a central question: does climate finance generate broader development outcomes as an externality? The assumption of positive development outcomes has justified billions of US dollars of financing, and there has been little evaluation of the impact of climate finance in this context. Using data from Fan, Wang, Zhong, Dong, and An (2025), I find that climate finance is associated with modest gains in private capital formation, a small increase in employment, and heterogeneous effects across labor force participation in addition to no observed relationship with health. These results are reflective in the estimated effect on GDP per capita growth and poverty measures. They also further suggest that climate finance may more greatly benefit countries vulnerable to climate change, but gender equity should be considered in its disbursement.

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Chapter 1

Introduction

The changing climate may be the most pressing and detrimental social issue facing our planet; though other impacts – such as the increasing numbers of natural disasters – draw greater public attention, the less visible effects of climate change on global poverty are also very consequential, particularly for the world's poorest (Dell, Jones, & Olken, 2012; Diffenbaugh & Burke, 2019; Gilli, Calcaterra, Emmerling, & Granella, 2024; Hallegatte, Fay, & Barbier, 2018; IPCC, 2022; Palagi, Coronese, Lamperti, & Roventini, 2022; Stern, 2007). Esther Duflo (2019) has pointed out that climate change "could easily undo all or most of the progress that has been made in fighting poverty," and research by Lankes, Soubeyran, and Stern (2022) and Bangalore et al. (2025) have reiterated this interdependence. Climatic impacts are already contributing to increased property losses from natural disasters, higher levels of malnutrition, work interruptions, and volatile food prices (Anderson & Thennakoon, 2015; Hallegatte et al., 2018; Lankes et al., 2022; Palagi et al., 2022). These consequences exacerbate the structural vulnerabilities that reinforce the poverty-inducing effects of climate change (Dell et al., 2012; Diffenbaugh & Burke, 2019; Gilli et al., 2024; Hallegatte et al., 2018; Hemming, Kell, & Mahfouz, 2002; Lankes et al., 2022; Palagi et al., 2022).

Given this context, the prospect of addressing the climate crisis through targeted investments has been described as the (most important) "development opportunity (of) this century" (Lankes et al., 2022). As Stern and Stiglitz (2023) and Ameli et al. (2021) note, credit constraints – partially driven by a high weighted average cost of capital – in the Global South are preventing optimal investment. However, this socially inefficient equilibrium can be disrupted through concessional financing that functions as subsidies. If this occurs, growth models that incorporate directed technical change suggest that these investments could improve market conditions for firms and reduce the growth drag of environmental externalities (Acemoglu, Aghion, Bursztyn, & Hemous, 2012; Hassett & Metcalf, 1995; Jaffe & Stavins, 1995). Furthermore, this exogenous spending shock may increase net job creation, particularly in recipient climate-related sectors (Batini et al., 2022; Bowen, 2012; Schwartz, Andrew, & Dragoiu, 2010; Wei et al., 2010). Lastly, appropriate projects sponsored by multilateral institutions may lead to positive health outcomes, stabilize food prices, and de-risk private investment (Anderson & Thennakoon, 2015; Foster & Rosenzweig, 1995; Ivanic & Martin, 2008; Mishra & Newhouse, 2009; Moctar, d'Hôtel Elodie, & Tristan, 2015; Mosley, Hudson, & Verschoor, 2004; Stern & Stiglitz, 2023). These arguments underscore the value of tackling poverty and the changing climate simultaneously in this manner.

Taking into account these arguments and the situation in the Global South, multilateral development banks (MBDs), multilateral climate funds (MCFs) and donor governments have decided to support projects around the world that claim to address climate change and poverty.[1] This support, defined as climate finance, can take the form of grants, concessional loans, guaranties, equity or other financial instruments intended to assist in adaptation and/or mitigation of the effects of climate change (Fan et al., 2025; UNFCC, n.d.; World Bank, n.d.).

Nonetheless, there is a lack of empirical evidence to support the claim that the tens of billions disbursed annually of climate finance fulfill their development aims (see Fig. 1). Arrticles typically examine climate finance in relation to trade, greenhouse gas emissions, donor lending preferences, or an economy's energy mix (Bayramoglu et al., 2023; Carfora & Scandurra, 2019; Liu & Paan, 2024; Rauniyar et al., 2025). There are other articles in the non-green foreign aid literature – containing similar financial instruments – that examine key poverty vectors such as capital formation and health, but this sort of aid lacks the climate benefits that would make it comparable (Alvi & Senbeta, 2012; Clemens et al., 2012; Hansen & Tarp, 2001; Mishra & Newhouse, 2009).

Fan et al. (2025) recently released a new dataset of MDB and MCF investments from 2000 to 2023. Given its release date, there are no scholarly articles utilizing it for a similar analysis, and it improves upon previous datasets in both its size and measurement. Moreover, considering MDBs and MCFs are the main components of (external) climate finance, it is a useful proxy for all international climate finance, including from bilateral sources Naran et al. (2025). Combined with other data sources, I use it to estimate the influence of climate finance on key vectors of poverty and poverty statistics; these

[1] It is important to note that donor countries have only partially fulfilled their commitments through their nationally determined contributions (World Resources Institute, 2025).

include output, the poverty headcount ratio, the poverty gap ratio, the squared poverty gap ratio, health, unemployment, labor force participation, and capital formation. The main model utilizes two-way fixed effects (TWFE) with clustered standard errors, and I use lags or accumulate climate finance commitments to examine the relationships between these variables across various time horizons. For each variable, I also investigate whether climate vulnerability and/or adaptation to climate change (measured with the ND-GAIN index) is an important factor in the relationship using an interaction term.

In chapter two, I draw on and add new components to existing theory – primarily representative agent models – to understand the mechanisms underpinning earlier claims. In chapter three, I discuss scholarly work on climate finance, foreign aid, and job creation to understand other key mechanisms and relevant methods of analysis. In chapter four, I explain in greater detail the data source and methods. In chapter five, I present my results, discuss them in relation to theory, and note the limitations of my analysis. Lastly, I consider how the results informs policy decisions and suggest possibilities for future research.

Chapter 2

Theory

Similar to typical foreign aid, I show that climate finance can impact poverty through demand shocks. However, as discussed by Rajan and Subramanian (2005), the purpose of aid is an important factor in dictating these impacts. I suggest that, in addition to these demand shocks, climate finance can further stimulate growth or directly impact those in poverty through the internalized externalities of climate change. These additional impacts distinguish climate finance from other categories foreign aid.

The following section is an intertwined discussion of both typical and climate-related channels. First, I propose that climate finance can impact poverty through an endogenous growth model following Chatterjee, Sakoulis, and Turnovsky (2003). I extend this framework beyond transfers by allowing for grants and debt accumulation among the public and private sector, incorporating transfers of climate-related technology, and considering the climate co-benefits of this investment. Second, I draw on New Keynesian theory to demonstrate that that climate finance – an aggregate demand shock – may potentially have further impacts on job creation, particularly for renewable energy investments (Harris, 2013; O'Callaghan, Yau, & Hepburn, 2022; Smets & Wouters, 2007). I then disaggregate labor market matching following Hall and Schulhofer-Wohl (2018) to further examine the sectoral effects from climate finance that complement the economy-wide effects originally modeled by Kopiec (2020). Lastly, I interpret the proposed drivers of poverty reduction (i.e. capital accumulation and job creation) through both a Post-Keynesian perspective (Dollar & Kraay, 2001) and a Neoclassical perspective (Barrett, Carter, & Chavas, 2018). By increasing capital stock accumulation and job growth, I hypothesize that climate finance can reduce poverty.

2.1 Capital Accumulation

I argue that climate finance can both increase spending through investment support and decrease depreciation by internalizing the externalities of climate change. Considering the complexity of the subsequent model, I begin with Rebelo (1991) to illustrate the importance of capital accumulation in overall output. Afterwards, I incorporate climate finance into a model by Chatterjee et al. (2003) and argue that it produces similar results. The last section of this chapter discusses the translation from capital accumulation to output to poverty reduction in detail.

2.1.1 Endogenous Growth

Rebelo (1991) constructs an endogenous growth model that emphasizes the importance of capital by intentionally keeping technology exogenous. While he later decomposes capital into two separate components, he begins by summarizing both physical and human capital with this variable.

Assume that the economy has two sectors of production: a capital sector and a consumption sector. The capital sector produces investment goods I_t,with some technology A, and the depreciation of the capital stock is standard at a rate δ. The consumption sector combines non reproducible factors with capital stock to produce consumption goods (C_t) using a Cobb-Douglass production function in which α is the output elasticity between capital stock, Z, and non reproducible factors, T. The model also assumes that the economy is composed of a set of representative agents defined as utility maximizing taking the form

$$U = \int_0^\infty e^{-\rho t} \frac{C_t^{1-\sigma}}{1-\sigma} dt. \tag{2.1}$$

where C_t is aggregate consumption, ρ is the subjective discount rate, σ is the coefficient of relative risk aversion, ρ models an agent's ability to have greater utility for a closer time horizon, and σ influences consumption depending on how risk adverse the agent is. These factors determine the steady state growth rate (g_y) expressed as

$$g_y = \alpha g_z = \alpha \frac{A - \delta_z - \rho}{1 - \alpha(1-\sigma)}. \tag{2.2}$$

where α is the elasticity of output with respect to total growth rate of capital, g_z. This model demonstrates the role of both physical and human capital in endogenous growth in a closed economy. Agent preferences dictate the total growth rate of capital, which has a strong influence on the growth rate, g_y. In an open economy, I suggest that external climate finance increases the level of public spending and decreases the depreciation rate by reducing damages from natural disasters, which directly influences g_z.

2.1.2 Integrating Climate Finance

Chatterjee et al. (2003) creates an endogenous growth model unilateral capital transfers (i.e. grants) that can increase government spending and capital accumulation. Chatterjee et al. (2003) is also able to model temporary transfers that are conditional on investment into public capital stock to estimate the effects on growth, net output, welfare, and balance of payments in the short and long run. I incorporate the disbursement of climate finance into this model by subjecting the economy to three separate types of (demand) shocks: i.) an investment specific technology shock, ii) a risk premium shock (motivating a one-time increase in private spending), and iii.) an exogenous (government) spending shock.

Consider a small open economy with infinite representative agents in the private sector who maximize utility subject to their budget constraint. Output is generated using the production function

$$Y = \alpha(\frac{K_g}{K})^{\eta} K = \alpha K_g^{\eta} K^{1-\eta}. \tag{2.3}$$

K can be interpreted as both stock of human and physical capital, where K_G is public capital, K is private capital, and α is some exogenous level of non-climate technology. I incorporate climate-related technology later in the evolution of investment and capital stock. Note that K and K_g are complementary; an increase in K_g has a large influence on the output by increasing the productivity of K.

Equation (2.3) further illustrates the importance of capital stock accumulation on net output, and it also illustrates the importance of government intervention in improving overall output through K_g. Instead of K_g to referring to public infrastructure spending (as in Chatterjee et al. (2003)), I redefine K_g as any public good or quasi-public good. Since climate finance spending is tied to a wide variety of private and public projects (and not only infrastructure), I make an important assumption to integrate climate finance into this framework (see Equation (2.4)).

I argue the validity of this assumption by demonstrating that climate finance spending – in both the public and private sector – is analogous in the manner initially presented in Equation (2.3). Climate finance deployed in the private sector directly translates into increased levels of investment in capital, K. Given incentives for profit-maximization, firms will likely abide by their loan commitments – avoiding legal penalties – and invest in K, improving output ceteris paribus. Furthermore, known credit constraints (Stern & Stiglitz, 2023) and the high cost of capital (Ameli et al., 2021) dis-incentivizes private sector investments of this sort in the laissez faire case. Concessional loans, grants, equity, and guarantees increase firms' capital availability, cost of capital, and perceived risk compared to the counterfactual.

Climate finance deployed in the public sector should also improve output by increasing K_G. A primary incentive of governments is continued reelection so, given these investments are publicly disclosed, they are similarly likely to abide by their commitments.[1] Additionally, these investments can be considered to be a quasi-public good – as in infrastructure – that are rival only to the degree of congestion, and various projects types supported by climate finance can introduce similar direct productivity effects – and can occasionally be non rival. This can include low-carbon transport, climate-resilient infrastructure, and nature-based solutions (Eguino et al., 2024). In this respect, there is sufficient justification – in addition to precedent as found in Barrow (1990) – for broadly defining the flow of government purchases (i.e. public spending) supported by climate finance to have complementary impacts on private productivity and output as in public infrastructure spending.

An additional dimension to consider is the climate co-benefits of these investments regardless of whether they are public or private. First, adaptation financing can address the increasing damage from natural disasters, which is a growing contributor of economic losses (see ch. 1, para. 1). Second, mitigation financing in renewable energy can increase air quality, which has known productive impacts (Chay & Greenstone, 2003). Third, the introduction of cheap renewable energy (relative to fossil fuels) (IRENA, 2025) can decrease the cost of production for firms, possibly encouraging further capital investments. Viewed in this light, climate finance is analogous in the productive nature of the spending, and it carries some additional productive benefits to existing capital not found in other multilateral or bilateral financing by addressing the increasing impacts of climate change.

I then denote CF to the be climate finance at time, t, zero, which can be decomposed as

[1]While I am unable to account for every climate finance project, public disclosure by MBDs and MCFs allowed Fan et al. (2025) to collect the dataset used in this study.

$$0 < CF = T_p + L_p + T_g + L_g. \tag{2.4}$$

Let T_p represent climate finance transfers (or grants) to the private sector, L_p the principal of climate finance loans to the private sector, T_g climate finance grants to the government, and L_g the principal climate finance loans to the government. The loan principal of climate finance contributes to the private debt, B, and government debt, A. For simplicity, I model climate finance as a one-time exogenous disbursement of some combination of transfers and loans from a donor – a MDB, MCF, and/or donor government – to the private sector and/or the recipient government. I restrict climate finance instruments to consider only loans and grants because i.) they represent a small share of total financing (see Figure 4.2) and ii.) this small share of equity and guarantee investments can arguably perform similarly to loans and grants in the manner by which they reallocate capital expenditure.

A key feature of the loan component of climate finance is the degree of concessionality of its interest rate as an incentive for the public and private sector to allocate resources towards these projects. This concessionality should decrease the average rate of interest for firms and the government. I express the private sector's interest rate, r_p ,as

$$r_p = \left(\frac{B_m r_m + B_c r_c}{B_m + B_c}\right) + \omega(N/K_T). \tag{2.5}$$

I let a portion of this debt be determined by the global market rate r_m and I introduce a separate, interest rate of concessional debt, r_c, where $r_c < r_m$ by definition to model the loan component of climate finance. Upon receiving the loan, total private debt B increases and the average interest rate r_w. I also allow for adjustments to the rate based on country-specific factors denoted as ω along with the national debt,N,which includes private and public debt, and total national capital K_T, which is inclusive of private and public capital. Similarly, I express the public sector's interest rate, r_g, as

$$r_g = \left(\frac{A_m r_m + A_c r_c}{A_m + A_c}\right) + \omega(N/K_T) \tag{2.6}$$

I express these separately since I expect that the risk premium is higher for firms, implying $r_p < r_g$.

An important fact of the nature of climate finance disbursement is that it is project-based (Naran et al., 2025). This can be expressed investments in private capital such that private capital stock accumulation can be defined as

$$\dot{K} = I_P + (k_2\varepsilon_t^i - \delta_K)K. \tag{2.7}$$

As discussed by Fan et al. (2025), MCFs conduct technology transfers and capacity building in the Global South; this captures some of the earlier discussion on climate co-benefits and further contributes to the evolution of capital stock. Let ε_t^i represent a disturbance to the investment specific climate technology[2] and k_2 denote the relative efficiency of installed capital compared to the overall investment effect. I define I_P as private investment and δ_K is the depreciation rate of private capital. Private investment can be further expressed as

$$I_P = \varepsilon_t^i[I_c + \delta_t(T_P + L_P)] \tag{2.8}$$

where I_c is the private sectors investment in areas other than climate finance and δ_t is the Dirac delta function. At any time, investment is chosen given a specified flow budget constraint, arbitrage relationships, and transversality conditions. At time, t, zero, the private sector is exposed to an exogenous shock of T_P and L_P, increasing their investments in K subject to marginal installation costs ($h_1(I/K)$). The (relevant) optimality conditions with respect to I_P and its arbitrage relationship with cost of borrowing abroad is described by

$$1 + h_1(I/K) = q = q'/\nu \tag{2.9}$$

$$\frac{(1-\tau)(1-\eta)\alpha K_g^{\eta} K^{-\eta}}{q} + \frac{\dot{q}}{q} + \frac{(q-1)^2}{2h_1 q} - \delta_K = r_p\left(\frac{N}{K_T}\right) \tag{2.10}$$

such that N is defined as the national debt, which is the principle and interest of all debts including

[2] Smets and Wouters (2007) create linearized DGSE model subject to an a variety of shocks, including exogenous spending shocks similar to climate finance. In the model, the authors conceptualize disturbances in investment specific technology to be a "first-order autoregressive process with an IID-Normal error term" in discrete time (p. 12). Given the model is in continuous time, I reinterpret this idea within my model such that ε_t^i can be further expressed as geometric Brownian motion that randomly evolves at some rate upon introduction at time 0. Given the size of MDB vs. MCF investment (composing most of the external aid provided of this class) (Naran et al., 2025), I do expect this effect to be small.

climate finance loans, and K_T is defined as total capital stock inclusive of private and public capital. I suggest that agents in the private sector optimize by allocating investment to climate finance initially because it temporarily increases the shadow value of the agent's private capital stock,q', by decreasing the cost of capital without influencing the unitary price of (publicly listed) foreign bonds, ν. This arbitrage relationship must hold with the lender(s) offering a concessional rate that decreases the r_p, increases N, and increases K such that it equates to the proposed adjustment in q' ceteris paribus; otherwise, the private sector does not accept the loan offer (and private investment does not increase).

Government spending, G, can be expressed as

$$G = \bar{G} + T_g \delta_t \tag{2.11}$$

where $\bar{G}$ is fiscal expenditure on public capital and CF_T is tied transfers as a result of climate finance. I restrict Equation (2.11) to only include tied transfers considering that a portion of climate finance is distributed as conditional grants and the nature of climate finance as an exogenous shock at time zero. I expected that $\bar{G}$ is tied to the scale of the economy so that

$$\bar{G} = \bar{g}Y + \delta_t L_g(N, K_T, Y) \text{ and } T_g = \sigma Y \quad \text{with} \quad \begin{cases} 0 < \bar{g} < 1, \\ \sigma > 0, \\ 0 < \bar{g} + \sigma < 1, \end{cases} \tag{2.12}$$

such that $\bar{g}$ is the rate of domestic government expenditure on K_g as a fraction of Y and, similarly, σ is the share of total output as climate finance grants/transfers. I incorporate concession loans from climate finance, L_g, into the model, and I specify CF_A to be a function of economy 's debt, total capital stock, and output in addition to L_g. I suggest that L_g is a rational choice by a donor to improve the chances of repayment by not burdening the economy with excessive debt. I expect that lenders would propose a smaller loan (or perhaps not introduce a loan at all) if if A/Y approaches a level exceeding 90%. This is because increasing debt would greatly harm the growth rate and the country's ability to repay such debt (Reinhart & Rogoff, 2010).

Equation (2.12) incorporates another important element of climate finance lending regarding the size of the financing through $\bar{g}$ and σ. As noted by Kaya and Leblebicioglu (2025), Galindo-Gutiérrez (2024), and Michaelowa and Michaelowa (2011a), the World Bank – and perhaps lenders in general – prefer not to lend to the least developed countries because middle income countries have stronger lending profiles (i.e. greater level of tax revenue, stronger government, and less climate risk). Noting this lending preference, the level of income is a convenient proxy for these factors and is an important factor for the model, and I extend this consideration L_g. A higher of G is inherently only capable for a larger economy with a lower risk, in which case it will receive a higher volume of lending.

The steady state in this model (without any modifications as specified above) finds that all real quantities grow at an equivalent constant rate at for the balanced growth rate of $\tilde{\phi}$. Given the complexity of the system, Chatterjee et al. (2003) conducts a numerical analysis of the original model to determine the impact of a tied productive temporary unilateral capital transfer of 5% of GDP. In the short run, they find an increased growth rate of private capital (1.74%), public capital (6.7%), and output (2.74%) – higher than that of a pure transfer. In the long run, there was an increase in the levels of total capital stock (8%), output (8%), consumption (8%), and welfare (5.32%).

Given the similarity of the financial instruments, I argue that the modified model above has similar effects to the original model with only transfers because of i.) the productivity of these investments more broadly, ii.) the added climate co-benefits, and iii.) the nation's ability to pay the debt. On the last point, given that these investments are productive, the below market rate of the loan component can be treated as a form of transfer, and lenders rationally choose a sustainable sum of climate finance debt (see Equation (2.12)), the nation is likely able to repay the loans introduced by climate finance. Furthermore, considering that climate finance should be multiple rather than single disbursements, I suggest that the individual impacts to capital stock should accumulate over the long run to produce similar beneficial effects.

2.2 Job Creation

Unlike other categories of foreign aid, climate finance primarily supports the renewable energy sector, thus creating new green jobs. It may also lead to secondary impacts across other sectors, which can include job creation and destruction. I begin by discussing Keynesian policy as a measure to stimulate the economy by addressing the costs of the climate crisis and elaborate on three moderating factors for the anticipated effects (Harris, 2013; O'Callaghan et al., 2022). I then reformulate both Kopiec (2020) and Hall and Schulhofer-Wohl (2018) to demonstrate that demand shocks as a result of climate finance can lead to improvements in employment in the short run during a recession.

2.2.1 Climate Finance and Keynesian Theory

I suggest that climate finance is a form of Green Keynesian policy – as proposed by Harris (2013) – that facilitates climate related investments. It induces an aggregate demand shock across the economy (as modeled in the previous section) that performs two key functions. In addition to the aforementioned capital accumulation effects, I suggest there should be additional employment and output effects.

I expect these effects to be moderated by three variables. To begin, the stage of the business cycle is an integral factor (Auerbach & Gorodnichenko, 2012; O'Callaghan et al., 2022). If deployed during a recession, climate-related investment can function as a counter-cyclical stabilization policy by utilizing idle labor capacity (Harris, 2013). However, even if this spending is deployed in such an upturn, I argue that there will be some effect on output.

Climate finance will also engender a structural transformation of the labor market to incorporate a larger renewable sector. In the short term, there will be lagged effects from structural and frictional unemployment. This will primarily depend on complementary labor market policies, and, if there is a sufficient increase in productivity from the technological improvements and the climate co-benefits of these investments, then it is reasonable to expect this second result (Smets & Wouters, 2007).

A final consideration is the heterogeneity of the investments across sectors. There is evidence to suggest that renewable energy investments, which compose a significant portion of climate finance (see Fig. 4.5), offer the greatest potential gains (Batini et al., 2022; O'Callaghan et al., 2022; Wei et al., 2010). Renewable energy projects – relative to fossil fuels – generate more direct jobs, create indirect jobs through energy cost savings that are reinvested into labor, and have high multiplier effects. These disaggregated impacts across sectors (including but not limited renewable energy projects) will ultimately cumulatively influence the overall net employment and output effects. I give these ideas a functional form – setting aside the stage of the business cycle – in the following section.

2.2.2 Employment Effects

Kopiec (2020) uses Keynesian ideas of fiscal stimulus to analyze the impacts on job creation in the short run with Bewley-Huggett-Aiyagari model to capture the price rigidities and then integrating idiosyncratic, endogenous unemployment risk into it. I suggest that climate finance should produce similar effects because labor market tightness is primarily linked to changes in aggregate demand (Braun, Bock, & DiCecio, 2009).

I begin by populating the model with four main actors: heterogeneous households (employed vs. unemployed), identical retailers, representative firms, and the government. These actors follow standard assumptions such as utility maximization, conditions for market clearing, and asset accumulation. The tightness of the labor market, χ, is expressed as

$$\chi = \frac{\upsilon}{1-(1-\sigma)n_{-1}} \quad \text{where} \quad \upsilon \in (0,10) \tag{2.13}$$

where υ is the replacement rate (i.e. unemployment benefits, etc.), σ is the separation rate, and the subscript n_{-1} indicates the previous period. Equation (2.13) is an expression of how easy is it for households to find a job or firms to find households to work. A higher χ indicates a tighter labor market and vice versa. If υ increases, the benefit of a household leaving the labor force increases. This decreases labor supply and increases competition among firms for households. If σ increases, then more households are leaving their jobs, increasing labor supply and decreasing competition among firms. When the labor market is tight, it may suggests that there is a higher supply of jobs; this may provide evidence to suggest it is the direct result of climate finance. The probability of finding a job is defined as a function of χ; this can be expressed as

$$f = f(\chi) = \frac{M(\upsilon, 1-(1-\sigma)n_{n-1}}{1-(1-\sigma)n_{-1}} = M(\chi, 1). \tag{2.14}$$

where M is some constant returns to scale matching technology. Given the earlier discussion on labor market tightness, it follows that the ability for firms and households to match is a direct result of χ.

I reinterpret Equation (2.13) and (2.14) to be heterogeneous across sectors. Following Hall and Schulhofer-Wohl (2018), I redefine matching as a function of tightness of each sector. Upon the disbursement of climate finance, I expect jobs to be created in the short run to fulfill immediate project needs. This implies a tighter labor market in the sectors of the disbursement of climate finance than on aggregate. I express this as

$$f_c = \gamma_c (V_c/H_c)^{\eta_c} \tag{2.15}$$

with f_c as the probability of finding a job in a sector stimulated by climate finance, γ_c as the matching efficiency of that sector, (V/H) as the ratio of vacancies to hires/the average duration of a vacancy for the sector, and η_c as the rate of conversion of job-seekers to hires for that sector. If disbursement reflects earlier assumptions of a one-time shock, I expect a simultaneous increase in labor demand across many sectors, particularly renewable energy, through direct employment by climate finance projects and indirect employment in other sectors benefiting from energy savings. This will increase in V_c across impacts these sectors. Furthermore, the aggregate tightness of the labor market, χ ,and probability of finding a job, f , should also directly increase from this effect in the short run.

Returning to Kopiec (2020), the amplification of this spending on labor market conditions is defined as

$$\sum_{\tilde{t}=1}^{+\infty} \frac{dC_t}{df_{\tilde{t}}} = \underbrace{\sum_{\tilde{t}=1}^{+\infty} \left[\sum_{a \in \{e,u\}} \int \frac{dC_t}{df_{\tilde{t}}} \left(b, z, \beta; \{\chi\}_{\tilde{t}}^{+\infty}\right) d\pi_a \right] df_{\tilde{t}}}_{\text{reduced unemployment risk channel}}$$
$$+ \underbrace{\sum_{\tilde{t}=1}^{+\infty} \left[\sum_{a \in \{e,u\}} \int c_{a,t} \left(b, z, \beta; \{\chi\}_{\tilde{t}}^{+\infty}\right) \left\{ \frac{d\tilde{\pi}_{\{a,t\}}}{df_{\tilde{t}}} \right\} db dz \right] df_{\tilde{t}}}_{\text{composition channel}} \tag{2.16}$$

such that C is aggregate consumption and π is the amount of employed agents in time t. For each agent, b is the asset holdings, z is a productivity level, and β is the discount factor. I interpret the reduced unemployment risk channel as the cumulative change in consumption with respect to a change in their probability of employment given their endowments and labor market tightness for employed, e, and unemployed, u, agents. If the chances of finding a job, f, increase, agents lessen their savings rate and increase their consumption, which will stimulates aggregate demand. This increase further improves output and labor demand. The consumption channel describes, given the aforementioned endogenous and exogenous factors, how changes in the probability of finding a job affect whether an agent is employed and aggregates this new higher level of consumption for all agents. This could also be interpreted as how consumption changes given that households have new income from a job, which has the same influence on demand, output, and labor demand. Depending on the multiplier effect of the project, initial job creation and consumption could induce additional consumption and investment.

It is these two channels that fuel a virtuous circle of improving labor market conditions following fiscal stimulus. Upon the disbursement of climate finance, which functions as an demand shock towards sectors related to climate change adaptation and mitigation, I expect similar effects. Tightness in labor markets within renewable energy, energy efficiency, and green infrastructure sectors can further stimulate output and labor demand in the short run. Continued investment over a long time horizon can lead to employment gains in these sectors.

However, it is important to note that these gains may also come at the cost of crowding out of jobs in other sectors; this idea is similar to the work of Autor, Dorn, and Hanson (2013). However, I find that crowding out through inter-industry competition is largely uncovered by the theoretical literature and out of the scope of this model.

2.3 Poverty Reduction

Given aforementioned theory, I hypothesize that climate finance increases capital stock accumulation and creates new jobs. I adapt Dollar and Kraay (2001) to suggest that it is reasonable for these variables to have meaningful effects on poverty. Adaptation financing is also argued to be an important factor in this relationship by mitigating climate damages.

2.3.1 From Growth to Poverty Reduction

In the empirical component of their paper, Dollar and Kraay (2001) find that no evidence to suggest that growth has disproportionate impacts on increases in income of the poor (which they define as the bottom fifth of the real income distribution). Instead, they suggest that incomes of the poor rise equiproportionally with average incomes. Furthermore, since there is no policy that can directly target the poor from a macroeconomic perspective, any sound growth enhancing policy also has a direct effect on reducing poverty.

Given earlier theoretical discussions, improvements in capital accumulation and labor market conditions should reduce poverty through output. Following Dollar and Kraay (2001), the above relationships can be described as

$$\underbrace{\frac{\partial Y_{ct}^{P}}{\partial X_{ct}}}_{\substack{\Delta \text{ in Net Income for Impoverished} \\ \text{from Growth Determinants}}} = \underbrace{\frac{\partial Y_{ct}}{\partial X_{ct}}}_{\Delta \text{ in Net Income}} + \underbrace{\left((\alpha_1 - 1)\frac{\partial Y_{ct}}{\partial Xct} + \alpha_2\right)}_{\text{Distributional Effects}} \tag{2.17}$$

$$\frac{\partial Y_{ct}^{P}}{\partial X_{ct}} = \underbrace{\frac{\partial Y_{ct}}{\partial J_{ct}}}_{\substack{\Delta \text{ in Net Income} \\ \text{from Job Creation}}} + \underbrace{\frac{\partial Y_{ct}}{\partial K_{ct}}}_{\substack{\Delta \text{ in Net Income} \\ \text{from Capital Formation}}} - \underbrace{\frac{\partial Y_{ct}}{\partial C_{ct}}}_{\substack{\Delta \text{ in Net Income} \\ \text{from Climate Damages}}} \tag{2.18}$$

where subscripts c and t indicate country and time, Y denotes net income, Y^P is the income of those in poverty, X refers to changes in growth determinants. The first term in Equation (2.17) is the effect on incomes of the poor of a change in one of the determinants of growth holding inequality constant, and the second term accounts for distributional effects. Within the second term, $(\alpha_1 - 1)\frac{\partial Y_{ct}}{\partial Xct}$ models the effects on the poorest part of the population through changes in inequality, where α_1 is estimated average income elasticity. The variable α_2 is any other direct effect of changes from growth determinants on incomes of the poor. Equation (2.18) decomposes the growth determinants into three sources: K is an increase in capital in the economy, J is an increase in the number of jobs, and C is climate damages. Assuming that there are changes in any of the growth determinants that positively influence output and there are no changes in equality, there should be a reduction in poverty. Some examples of these changes include an increase in the amount of jobs (increasing employment), a greater level of gross fixed capital formation, and a reduction in climate damages. Furthermore, if adaptation financing is implemented, climate damages – which affect the poor disproportionately (see Ch. 1, para 1) – will be internalized or mitigated, further improving income for those in poverty. Considering the above, I hypothesize that increases in capital stock accumulation, job creation, and direct adaptation financing – all of which are the result of climate finance – should also decrease poverty.

2.4 Conclusion

Theory indicates that climate finance can influence poverty through two key vectors: capital accumulation and job creation. Increases in the level of climate finance should influence production through capital accumulation, and I argue that quasi-public nature of the investment, the climate co-benefits, and likely ability for a recipient nation to pay off its debt allow my adjusted model to have the same effect as in the original model by Chatterjee et al. (2003). Keynesian theory suggests that there should also be short and (potentially) long run job gains from fiscal stimulus incorporating climate goals depending on the stage of the economy's business cycle, the composition of climate finance across sectors, and the size of the multiplier effect (Hall & Schulhofer-Wohl, 2018; Harris, 2013; Kopiec, 2020; O'Callaghan et al., 2022). Assuming the viability of these poverty vectors, capital accumulation and job creation may potentially influence poverty by through induced changes in average output assuming no changes in inequality (Dollar & Kraay, 2001). Adaptation financing may also directly improve the incomes of the poor as they are the most affected by climate damages.

Chapter 3

Literature Review

It is clear that there is theoretical evidence to suggest that climate finance has a strong relationship with key vectors of poverty, and the empirical evidence also supports these conclusions. This chapter reviews the approaches of relevant papers supporting this hypothesis. Additionally, it analyzes the methodologies used to evaluate such questions, and health is introduced as an important vector of poverty associated with climate finance.

3.1 Climate Finance

Considering they are the most relevant, I begin by discussing an article by Bayramoglu et al. (2023) and Zhao et al. (2025) on the effect of climate finance on trade and the labor force participation. Bayramoglu et al.'s (2023) methods are broadly representative of the limited bilateral climate finance literature (Carfora & Scandurra, 2019; Lee, Li, Yu, & Zhao, 2022; Liu & Paan, 2024), and Zhao et al.'s (2025) findings provide additional context for the results of this thesis.

3.1.1 Bayramoglu et al. (2023)

There are not many articles studying international climate finance in this context. While the mainstream literature largely focuses on risk-propagation (Giglio, Kelly, & Stroebel, 2021), papers studying development impacts of climate finance remain understudied beyond its influence on carbon emissions (Carfora & Scandurra, 2019; Lee et al., 2022; Liu & Paan, 2024). Bayramoglu et al. (2023) finds evidence that there is a relationship between bilateral trade and bilateral official development assistance (ODA) climate finance allocation from the same donor countries. Although this study will focus on multilateral climate finance, the authors' methods are highly relevant and shall be the main focus on this review.

Context and Motivation

The authors suggest that bilateral climate finance allocation is motivated not only by donor need but also by donor interests, the latter of which the authors seek to estimate. By allocating climate finance to these countries, donor countries seek to maintain colonial trade relations (degraded by climate change), improve imports to the donor countries by reducing the price of goods, and increase the income of the former colonial country, which increases exports from the donor country.

Data, Methods, and Results

The authors use the OECD-DAC statistics for bilateral aid. These commitments (and not disbursements) are classified via the Rio markers to indicate how relevant the project is in addressing climate change, excluding projects that are Not Targeted. Additionally, they aggregate both adaptation and mitigation aid and screen for duplicate transfers. No transfers were recorded as zero. The authors also use the

CEPII BACI database for trade, and their control variables can be broadly classified by recipient need, recipient merit, past colonial ties, and donor interests. Lastly, all the monetary variables were corrected for inflation using the 2010 US CPI.

The baseline model is a ordinary least squares (OLS) fixed effects (FE) regression specified as

$$\text{Climate Finance}_{ijt} = \alpha_X Exports_{ijt} + \alpha_M Imports_{ijt} + \beta Z_{ij} + \theta Y_{itj} + \gamma W_{it} + \nu X_j t + FE + \varepsilon_{ijt} \tag{3.1}$$

where Z_{ij} is bilateral time-invariant control variables, Y_{ijt} is time-varying control variables, W_{it} is donor-year controls, and X_{jt} is recipient year controls. Two-dimensional FE including donor-year and recipient-year FE are used for quasi-within-estimation.

However, there are endogeneity issues with the regression wherein there is simultaneous determination of climate finance and trade. To address this, the authors employ an instrumental variables (IV) approach using shift-share instruments of exports and imports at the bilateral-year level. The instrument is found to have a high F-statistic and has relevancy.

The main result is that donor exports have a significant effect (and causal impact) on climate aid allocation. As robustness checks, the authors employ alternative instruments and find similar results. Additionally, to show the results are robust corrected for sample selection (since aid flows are only positive), the authors employ a Heckman correction. Additionally, since the estimates do not largely differ between the IV and OLS, endogeneity is assumed to be minor and the OLS estimates are preferred.

Analysis

Bayramoglu et al. (2023) document a strong theoretical and empirical relationship between trade and climate finance. Additionally, they employ a strong instrument and many robustness checks on key results. Their method of addressing endogeneity is relevant as I expect to be dealing with endogeneity issues of a similar sort. In particular, I expect donor preferences and recipient needs to be major issues in attempting to estimate the causal effect of climate finance on poverty vectors (Bandyopadhyay & Wall, 2006).

I am concerned about the data the article employs. The OECD-DAC data has been documented to be inconsistent with the reliability of documenting "true" climate finance (Michaelowa & Michaelowa, 2011b). In other words, there are many projects documented as climate aid that are misreported and have little to do with addressing climate change. I address this concern and take inspiration from Bayramoglu et al.'s (2023) empirics in my analysis.

3.1.2 Zhao et al. (2025)

There is little literature available on the distributional consequences of climate finance on development indicators. Zhao et al. (2025) examines this in the context of labor force participation across men and women with two way fixed effects (TWFE). The authors find there are differential impacts, and they explore possible explanations with additional regression specifications. These findings complement my results.

Context and Motivation

As the main purpose of climate finance is improving equity on a country level, the authors argue that this should extend to vulnerable groups affected by climate change, primarily women. They claim that structural inequalities largely prevent women from obtaining the benefits of climate finance, and paying considerations to equity in its distribution is important in improving women's resilience to the impacts of climate change. Therefore, it is important to explore the determinants that contribute to possible inequality and labor market impacts across gender.

Data, Methods, and Results

The authors use the OECD-DAC statistics as their main dataset for climate finance, which contains both multilateral and bilateral climate finance received. However, they authors do not provide enough clarity to ascertain the exact data source. The likely source is Climate Related Development Finance Dataset (CRDF), which contains project level data by sector with Rio markers. Data with Rio markers is problematic for aforementioned reasons discussed in the previous subsection. Unlike Bayramoglu et al. (2023), I assume that they include all Rio marked aid, included aid that is Not Targeted. Furthermore, the authors define climate finance observed at a country level "received" in the article, but, in fact, the

CRDF data is recorded as commitments. This is clearly misleading. Additionally, the authors (natural) log transform climate finance, presumably to adjust for skewness.

For female labor force participation, they use data from the international labor organization (ILO). They define this as "share of the female population (age 15 and above) who provide labor to provide and provide goods and services" (Zhao et al., 2025). Furthermore, they include ILO imputed data in addition to nationally reported data.

With regards to their methods, the authors use TWFE. It is unclear whether clustered standard errors are employed, and the authors exclude observations 2% in size before and after the explanatory variables. To examine heterogeneity across sectors, the first and main set of regression specifications the authors use climate finance disaggregated by sector and country-level macroeconomic controls. These controls are presumably employed across remaining specifications, and the authors lag climate finance one period, remove BRICS countries, and use a the natural log of the three-year mean of climate finance as a robustness check. Afterwards, further subgroup analysis is done at the country level (region, income group, Economic Risk Index, Financial Risk Index, World Uncertainty Index), by women group (region, ethnic fractionalization, proportion of seats held by women in national parliaments, fertility rate), and by a country's gender norms.[1] There is also an additional set specifications to see the impact climate finance (aggregate, mitigation, and adaptation) on female and male labor force participation across sectors.

The authors find a decrease in female labor force participation. This is mainly is associated with mitigation financing, and these results persist across the robustness checks. Across countries, the author find a statistically significant and larger, negative effect for climate finance distributed to countries with high Economic Risk, Financial Risk, and World Uncertainty indexes in addition to high fertility. Furthermore, while the coefficient is negative for low to low-middle income countries, there is a positive coefficient for upper-middle income countries, and gender norms with regard to female-employment and contributing to family work are significant when interacted with climate finance. Lastly, climate finance is associated with a broad decrease in participation by women in the service sector, largely associated with mitigation financing, and an increase in men in the industrial sector, which is associated with adaptation and mitigation financing.

[1] These indexes are from the ICRG dataset.

Discussion

The authors suggests a few explanations for the main finding. First, women may be less likely to find a job if household incomes improve as there are less economic difficulties. Second, social norms may be driving these findings as projects are not designed to address distribution across gender. For example, women's freedom from housework or access to education may constrain them from being employed in climate finance projects, and the authors find that climate finance is typically deployed in male dominated industries.

For the other results, the authors suggest that countries bad macroeconomic environment will i.) reflect this in their treatment of vulnerable groups and ii.) any instability and uncertainty will lead to issues in the distributing of financing. Additionally, countries with high fertility may lead women to have time and resource constraints the prevent them from entering the labor force, and gender norms will prevent women from accessing climate finance-related jobs and resources. Lastly, given this context, women entering the labor force may crowd out women by exacerbating these gender-related distributional factors.

Analysis

Zhao et al. (2025) create a compelling narrative of differing factors driving the decrease in female labor force participation in response to climate finance. The heterogeneity analysis was detailed, and I their robustness checks were sufficient. However, I have concerns with transparency in the documentation of their data sourcing and choice of eliminating outliers. Additionally, while TWFE is suggestive, other models should be used to complement it to ensure the validity of results. In this article, only TWFE are employed because of time constraints.

3.1.3 Synthesis

Bayramoglu et al. (2023) and Zhao et al. (2025) are strong examples of articles that analyze the externalities of climate finance. Bayramoglu et al. (2023) has strong methodology and robustness checks, and Zhao et al. (2025) has strong explanations for decreasing labor force participation by women after the introduction of climate finance. I find that both articles draw from the same data source, which has issues in the reporting of climate finance. I account for this in this thesis.

3.2 Job Creation

In the previous chapter, work by Harris (2013), O'Callaghan et al. (2022), and Kopiec (2020) is used to suggest that climate finance functions as an aggregate demand shock, increasing job creation. This employment effect is suggested as a key mechanism that could reduce poverty (Barrett et al., 2018; Dollar & Kraay, 2001). This section will briefly review key empirical evidence to support green fiscal stimulus that supports job creation in the energy sector. While climate finance not only supports projects in this sector, the following evidence is argued to more broadly apply to climate finance spending.

3.2.1 Batini et al. (2022)

To create jobs, fiscal expenditure first needs to have a fiscal multiplier greater than one. If the multiplier is less than one, then a dollar spent by fiscal stimulus will have less than a dollar impact on GDP (Batini et al., 2022). This would negate any potential positive effects on job creation as in Kopiec (2020). Therefore, it is imperative to understand the size of the multiplier in general. Batini et al. (2022) estimates a multiplier effect above one; this may serve as a reference point for climate finance spending in general.

Context and Motivation

Batini et al. (2022) seeks to determine whether there is a tradeoff (as in Brock and Taylor (2010)) or synergetic relationship between "green spending" — spending on renewable energy and biodiversity — and the economy. This was particularly salient in the context of the COVID-19 pandemic when many fiscal stimulus packages were being disbursed. To estimate the multiplier between green spending and GDP, the authors calculate the investment multiplier using data from several public and non-public sources.

Methods, Results, and Robustness Checks

Batini et al. (2022) first begin by computing the multiplier for various spending categories using panel VAR models of the form

$$y_{it} = \rho_i + \gamma_i + A^1 y_{i,t-1} + ... + A^P y_{t-p} + B_i x_{i,t} + \varepsilon_{i,t} \tag{3.2}$$

where t denotes the time dimension, i is the country dimension, and p is the lag structure. y is the vector of endogenous variables, x denote exogenous variables (for some specification, A^1 to A^P are dynamic coefficients for endogenous variables, B_i are coefficients for the exogenous variable, ρ_i is country FE, γ_i denotes time FE, and $\varepsilon_{i,t}$ is the vector of normally distributed residuals. The model is then calibrated using a Bayesian approach to assemble the posterior distribution, which is used to generate an impulse response function over five years. Cumulative multipliers are then derived from the impulse response function by dividing the change in GDP by the change in spending on energy.

In response to a shock (or investment) of 1% of GDP, the multiplier effect for renewable energy is 1.19 compared to 0.65 for non-eco-friendly energy investments immediately after the shock (Batini et al., 2022). In five years, the multiplier effects are 1.11 and 0.52 for renewable energy and non-eco-friendly energy respectively. To verify the robustness of the results, the authors adjust the lag structure, apply a separate filter to compute real GDP estimates, and try alternative specifications.

Analysis

In terms of results, it is important first note that these multipliers are likely an underestimate of the true multiplier. This is because Batini et al. (2022) is only estimating the direct impacts and not indirect effects. Yet, the result is promising in demonstrating the positive economic impact of investments in clean energy.

3.2.2 Wei et al. (2010)

Wei et al. (2010) focuses on different energy technologies (renewable, carbon sequestration, and energy efficiency improving) can generate jobs in the US economy as modeled in Ch. 2. Given a large component of the climate finance in the study sample is for renewable energy projects (see Fig. 4.5), this articles illustrates its potential effects. Although its projections are limited to the US economy, the underlying mechanisms for these projections that are found in the study may be generalizable to other economies. Additionally, while this study looks at the impacts of the renewable energy and biodiversity spending, the former will be the only component reviewed below.

Context and Motivation

Wei et al. (2010) notes that other studies have found economic benefits of renewable energy through job creation. Using 15 studies with varying results of job creation through energy production and carbon sequestration, the authors create an analytical model to create a "meta study" by taking ranges and averages of job multipliers of potential job creation in the USA from non-fossil fuel technologies. These technologies include are either renewable energy, energy efficient or low carbon. Unlike previous work, the authors also attempt to account for indirect job losses as a result of the creation of clean energy jobs to find the net employment impacts. For example, an indirect effect may be a decease in the number of jobs in the fossil fuel industry as a result of the introduction of renewable energy.

Methods

Largely, these studies are reports by NGOs, national laboratories, and universities, and a few are from peer-reviewed journals (Wei et al., 2010). Their estimates of job creation range widely, focus on the developed world, and are either analytical-based studies or input output (I/O) models. These estimates are normalized to create a measure called "lifetime average employment per unit of energy." Additionally, to compare technologies with different capacity, employment per unit of energy is measured with "job-years per GWh" or "job-years per average MW." The authors note that jobs created can separated into two groups: i.) construction, installation, and manufacturing and ii.) operations, maintenance, and fuel processing. These are then separated by study for comparison. It is also important to note that, if an improvement in technology directly improves energy efficiency, there will be "induced" job creation by increasing household savings, and there may also be indirect jobs from the inputs and outputs of these projects.

The authors then discuss their analytical model (Wei et al., 2010). Fundamentally, this model depends on the earlier normalization of the data and macroeconomic assumptions. For example, they assume reductions in electricity demand are from energy efficiency and not the result of reduced energy usage from behavioral changes.

Results

There are three key results to this paper. First, Wei et al. (2010) demonstrate through projections that increasing energy savings increases the (net) amount of jobs created in the economy. Second, increasing renewable portfolio standards (the amount of renewable energy in the energy mix) in addition to improvements in energy is associated with a high amount of (net) job creation in the entire economy. A similar result is replicated for both the carbon capture & sequestration and the nuclear energy sectors. However, in each individual sector, high improvements in energy efficiency will decrease the amount of jobs in that sector. Lastly, the authors note that, if geothermal, solar, or solar thermal targets are higher than in the business-as-usual case, there will be large increases in job generation since they have the highest job multipliers across the reviewed studies.

Analysis

Wei et al. (2010) provides an insightful analysis that illustrates some key mechanisms for how different energy technologies may create jobs in a variety of alternative manners. Using various studies, the authors demonstrate different scenarios that show sizable projections for job creation. In these projections, a key point is that, although data is based on a variety of stakeholders, these stakeholders may hold different biases in their estimates, with some types technologies having very few studies. By accumulating all of these studies, the authors can address these biases to determine the net employment impacts. They also consider how improvements in energy efficiency can also drive job creation beyond direct employment.

3.2.3 Synthesis

Wei et al. (2010) and Batini et al. (2022) provide evidence to suggest that investments in renewable energy can create jobs. Batini et al. (2022) proves fiscal expenditure on renewables has a multiplier greater than one. Wei et al. (2010) uses an analytical model using data from 15 other studies to demonstrate these investments also can create jobs though direct employment at the project, indirectly via the projects inputs or outputs, or induce other jobs through energy efficiency improvements.

More broadly, it can be argued that the key principles that make renewable energy investments create a high degree of jobs can be applied to climate finance more broadly. According to Batini et al. (2022) and Keynes (1936), the component to a high multiplier effect are investments with high domestic content that are labor intensive, and the investment is then spent in the economy, creating a cyclical effect. As is important in Wei et al. (2010), many of climate finance projects (at least with the sample data) would need with high domestic content and are labor intensive (see Figure 2 in the following chapter). For example, investments in transport and environmental improvement would need to use some amount of domestic labor. It is possible that, while there is evidence to suggest renewable energy may be the most potent category, other sorts of climate finance spending may also have a positive effect on GDP and job creation. However, as is mentioned by O'Callaghan et al. (2022), the second element of this cycle critically depends on the state of the business cycle for all investments. This will be considered in the following Chapter.

3.3 Foreign Aid

Climate finance (in the ODA literature) utilizes the same concessional financing structure as foreign aid more broadly (International Development Association, n.d.; OECD, n.d.).[2] This chapter will review some key articles from the foreign aid literature that support theoretical predictions about the effect of climate finance. These articles discuss the effect of foreign aid on capital accumulation, growth, poverty (measures), and health outcomes (Clemens et al., 2012; Hansen & Tarp, 2001; Mishra & Newhouse, 2009; Mosley et al., 2004).

3.3.1 Hansen and Tarp (2001)

Theory (Chatterjee et al., 2003; Rebelo, 1991) suggests that capital accumulation is a key factor in increasing income and, as a result, reducing poverty. Hansen and Tarp (2001) provides empirical evidence for the first premise of this hypothesis. Additionally, the authors suggest that capital accumulation is a key mechanism in this relationship.

[2]There is separate strain of literature that utilizes the term climate finance to refer to the integration of climate risk in macrofinance models (Giglio et al., 2021). To be clear, in this study, I am referring to climate finance as the disbursement of concessional financing instruments as other ODA (or foreign aid) articles.

Context and Motivation

Hansen and Tarp (2001) begin by discussing the varied results of the long literature on the subject of foreign aid and economic growth. They notably discuss the Burnside and Dollar (2000) result of the threshold effects of aid and a government policy interaction term, and the constraint of good government policy in allowing aid to work effectively. They also discuss other articles that introduce the idea of decreasing marginal return to aid by squaring the aid variable in the regression. Lastly, they note major concerns about endogeneity. In a separate paper, Rajan and Subramanian (2005) assert that this noise in the same or similar datasets leads “flimsiness” in the estimates of most articles. Rajan and Subramanian (2005) claims this is the sole contributor of articles the many contradictory results of articles on this subject. Hansen and Tarp (2001) remain firm that their estimates are not “flimsy” and find a positive impact of aid on growth. Additionally, they provide empirical evidence to support that the aid-growth relation through investment and human capital.

Aid and Growth

Hansen and Tarp (2001) begin by constructing a policy index to serve as a the policy variable as in Burnside and Dollar (2000). This index is a combination of a budget surplus (weighted the greatest), inflation, and openness to trade. Afterwards, they use a series alternating OLS regressions with aid, aid squared, and an aid-policy interaction term. Their complete set of controls include budget surplus, inflation, openness, financial depth, ethnic fractionalization, assassinations, institutional quality, and initial GDP per capita. They find positive significant results that aid is positively associated with growth across all specifications, but they demonstrate the main result of Burnside and Dollar (2000) — i.e. the significance of aid-policy interaction — is only replicated after removing outliers from the sample.

Hansen and Tarp (2001) also discuss simultaneity bias. They conclude that, while Durbin-Wu-Hausman tests in previous studies do not find endogeneity on OLS and IV estimates, they suggest that both estimators are inconsistent and not tests of the endogeneity of aid. Since their previous results probably have endogenous variables and need to include country-specific effects, they proceed to use a generalized method of moments (GMM) approach. This improves upon a standard FE model that requires exogeneity of the covariates with respect to $\varepsilon_{i,t}$, and, while some information is lost in differencing, the number of countries accounts for this. The authors also test for serial correlation through a Sargan test, and test various lags, which leads the authors to exclude Budget Surplus and Financial depth from the final regression. Each regression finds significant effects on the growth rate from aid, differing from the OLS and IV estimates. This suggests improvements in accounting for endogeneity.

Capital Accumulation

To analyze whether aid affects growth because of investment, they rerun the growth regression to include investment and human capital through various OLS, FE, and GMM regressions. In the first, they conduct OLS and GMM regressions for aid and growth, where they attempt to account for all sources of capital accumulation by including gross domestic investment, FDI, and a measure of human capital. By varying the inclusion of these variables (adjusting the coefficients in their regressions), they find results suggestive to the fact that investment is the main contributor to growth excluding highly aid-dependent countries. This result is bolstered by a FE and GMM regressions of aid on investments controlling for capital accumulation, where FDI crowds out domestic investment. The authors observe endogeneity by comparing both regressions, and the results suggest that aid is positively associated with domestic investment.

Analysis

Hansen and Tarp (2001) conduct a rigorous assessment of aid and growth, incorporating new insights on the underlying mechanism. They build on the endogenous growth literature (Rebelo, 1991; Solow, 1956) by suggesting that foreign aid contributes to through growth through human capital accumulation measured through different sources of investment. Even so, their methodology has some flaws. For example, their choice of instrument and use of GMM regressions may have be inappropriate considering that past aid would have an effect on current growth. Additionally, a more comprehensive set of controls with indicators for health and education would have removed bias from the estimates. Issues with similar methods will be discussed in the following review.

3.3.2 Clemens et al. (2012)

Building from earlier articles (including Hansen and Tarp (2001)), Clemens et al. (2012) provides a more sound empirical strategy to the evaluation of aid on growth. These notes on econometric design and methodological issues are relevant in the choice of this study's model specification and robustness checks.

Context and Motivation

Clemens et al. (2012) attempt to write the definitive article on the subject of foreign aid and growth by extensively reviewing and improving on the large literature on the subject of aid and economic growth. Clemens et al. (2012) concurs with Rajan and Subramanian (2005) by suggesting that the instruments of previous studies required to deal with the endogeneity have been of "questionable" validity and strength such as lagged added, political ties of donors, and population size. Additionally, Clemens et al. (2012) also suggests that the varied results are from the "the timing of causal relationships between aid and growth." In other words, some projects funded by aid have impacts much sooner than other projects. The authors provide a simple, standardized approach that remarkably provides similar results across all papers that addresses these issues.

Notes on Timing and Identification

If positive effects on aid are not on growth are also not observed, then Clemens et al. (2012) suggests that it is because they chose to examine aid that is not expected to produce results within the time period of the panel. However, opting to use high levels of observations for each period to capture these effects requires the use of cross-sectional estimators with limited degrees of freedom, reverse causation, and simultaneity bias from omitted variables. Therefore, the authors use short periods to decease omitted variable bias and difference away country FE. Additionally, the authors disaggregate aid to only include "early-impact" aid. This means any projects that affect infrastructure, communications, energy, banking, agriculture, industry, and any ambiguously classified aid.

Results and Robustness Checks

Clemens et al. (2012) alters the regression design of the three most cited papers (at that moment) in the aid and growth literature. The only change Clemens et al. (2012) makes is i.) using lagging and differencing to avoid poor quality instruments as mentioned above and ii.) disaggregating aid after replicating results . The authors claim that all other elements of the regressions — both OLS and IV

— are the same. The authors finds the results are remarkably very similar with a consistent instrument of purported higher quality. They also make an important note that inducing initial GDP per capita like Burnside and Dollar (2000) and Hansen and Tarp (2001) may introduce dynamic panel bias.

Clemens et al. (2012) also conducts a number of robustness checks. To address possible mean reversion and reverse causality, the authors take the most representative regression and twice-lag the growth variable; the coefficients do not substantially change. The authors also adjust a main variable of interest, (estimated) early impact aid, with a separate similar variable, finding no substantial difference. Lastly, the authors take issue with some aid-growth regressions that drop outliers that heavily influence the results and acknowledge colinearity between aid and the aid squared term. This is addressed using a semi-parametric analysis.

Analysis

Clemens et al. (2012) conducts a thorough examination of the methods of the aid-growth literature and extensively addresses issues with previous papers. The authors carefully elaborate on the issues of each paper they discuss, and they also demonstrate the many flaws with the IV and GMM regressions of previous papers. These observations will be heavily incorporated into the methods of this paper.

3.3.3 Alvi and Senbeta (2012)

Although Hansen and Tarp (2001) and Clemens et al. (2012) were relevant for their insight into the growth-aid literature and methodologies in relation to this study's theory, it is also instructive to examine the effect of foreign aid on poverty directly rather than focus on the intermediary variable of output. Fundamentally, the literature finds that the share of government spending on "pro-poor expenditure" as a result of foreign aid is the key driver of reductions in poverty (Mosley et al., 2004). Alvi and Senbeta (2012) concurs with this view, but the authors estimate this relationship using total aid and disaggregating by type.

Context and Motivation

Alvi and Senbeta (2012) begin by claiming that foreign aid decreases poverty through two main channels. First, the authors acknowledge that foreign aid decreases poverty by increasing average income. By increasing average incomes, a secondary effect is decreasing poverty assuming no changes in inequality (similar to Dollar and Kraay (2001) and Ravallion (2022)). Second, the authors strongly suggest that "pro-poor" foreign aid can directly affect poverty if targeted to low-income households.

Alvi and Senbeta (2012) also discuss disaggregating aid as a method to shed light on underlying mechanisms. The authors contest that bilateral aid is ineffective because i.) it is disbursed with strategic/geopolitical motivations and ii.) it lacks the technical support of a multilateral agency. Additionally, the authors suggest that loans may be ineffective at reducing poverty since they must be used for some "financially productive purpose" compared to grants (Alvi & Senbeta, 2012).

Methods

Alvi and Senbeta (2012) estimate the effects of aid on poverty by loosely drawing from the idea of a growth elasticity of poverty (Dollar & Kraay, 2001; Ravallion, 2022). The regression equation is specified as

$$\log P_{it} = \alpha \log P_{i,t-1} + \beta_1 \log Y_{it} + \beta_2 \log G_{it} + \beta_3 AID_{it} + X_{it}' \theta + v_i + \varepsilon_{it} \tag{3.3}$$

where P is a measure of poverty (headcount index, poverty gap, and squared poverty gap), Y is real income per capita, G is the Gini coefficient, AID is aid flows, X is a vector of other policy and institutional variables that could affect poverty, and the subscripts i and t refer to country and time respectively. The lagged effect of poverty is used to incorporate the persistent effects of poverty; this requires that a GMM estimator is utilized.

Results and Robustness Checks

Alvi and Senbeta's (2012) first relevant finding is that, across all measures of poverty, there is a negative, significant relationship between aid flows and poverty controlling for income and other factors. In the same regression, income, inequality, and lagged poverty mores were also significant factors. Next, when disaggregated by bilateral and multilateral aid, the authors find that only multilateral aid produces the poverty-reducing effects. Finally, if disaggregated between grants and loans, the authors only find significant effects of grants on reducing poverty (headcount index only). However, on this last finding, it is important to note that the coefficients on the loan variable were negative and had relatively low p-values (approximately 20% across specifications).

To validate instruments and moment conditions, Alvi and Senbeta (2012) note that they use a Sargan test and Arellano-Bond first-order and second-order serial autocorrelation tests as robustness checks. They find that their instruments are valid, and there is no second-order autocorrelation. The authors also conduct OLS estimates to compare to the GMM estimates, and they have specifications in each with and without control variables. It is further important to note that outliers are excluded, but the authors are note specific about their exclusion criteria.

Analysis

Given that the data is expected to be noisy (as in Hansen and Tarp (2001) and Clemens et al. (2012)), the methodology chosen by Alvi and Senbeta (2012) is a critical component of this paper. Mainly, the use of Difference GMM — similar to how it was utilized in Clemens et al. (2012) — would have been helpful additional robustness check to verify the results. Additionally, verifying the strength of the instruments through an F-statistics would confirm that they are not weak. Lastly, the exclusion of outliers is an important but rarely mentioned note in the paper. This may not only make the results of this paper hard to replicate but also significantly impact results.

Alvi and Senbeta's (2012) disaggregation of aid also reveal some important relevant empirical evidence. Considering the sample of climate finance in the paper is multilateral, this does demonstrate some support for the expectation of significant, negative effects. However, it simultaneously discredits the earlier assumption about Chatterjee et al.'s (2003) work that grants and loans produce similar effects. To this point, this study points to two key facts. First, the disaggregation of grants and loans largely reduces it statistical power. It is possible the results are a Type II error. Second, the loan regressions consistently produces estimates with low-p values near the 10% threshold and had negative coefficients as expected. Therefore, while not as effective as grants, loans should be similarly effective.

3.3.4 Mishra and Newhouse (2009)

The bi-directional relationship between poverty and health is well established in health economics (Grossman, 1972; Lleras-Muney, Schwandt, & Wherry, 2025). In essence, poverty is associated with poor health outcomes because it can lead to factors that diminish health (psychological stress, poor nutrition, exposure to environmental hazards), and poor health decreases productivity and increases medical expenses, preventing investments in better health (Lleras-Muney et al., 2025). Given infant mortality's high degree of association with income (Waldmann, 1992), Mishra and Newhouse's (2009) contributions are highly relevant to this review, particularly in relation to my methods and results.

Context and Motivation

Foreign aid (defined as the sum of grants and concessionary loans) has been credited in some cases to improve health outcomes by reducing resource constraints and improving health service delivery (Mishra & Newhouse, 2009). Since foreign aid and infant mortality are closely linked, Mishra and Newhouse (2009) believe that it is easier to detect a relationship between these variables compared to an estimation between aid and growth. Additionally, Mishra and Newhouse (2009) sheds light on broader questions about why only health aid may be effective in improving health outcomes and suggests a potential mechanism.

Methods

Mishra and Newhouse (2009) generally compare two different types of regressions in the following results. First, the authors use an OLS estimator of the form

$$\log IM_{rt} = \alpha \log A_{rt-1} + \gamma \log IM_{rt-1} + \beta X_{rt-1} + \delta_1 HIV_{rt} + \delta_2 W_{rt} + \nu_t + \varepsilon_{rt} \tag{3.4}$$

where IM is infant mortality, A is aid per capita, X is the vector of control variables, HIV is a scalar representing the incidence of HIV/AIDS, W is the presence of war, and the r and t subscripts refer to recipient country r in time t. A and IM are in log-log form in line with previous literature to smooth the data and adjust for initial levels of variables.

Mishra and Newhouse (2009) then notes that the OLS may be biased because of endogeneity and measurement error. Additionally, the FE regressions would be further biased because of time-varying unobserved characteristics in the error term, the within-estimators lagged/predetermined variables being inconsistent, and measurement error biasing coefficient towards zero. To address this, the authors use a system GMM of the form

$$\log IM_{rt} = \alpha \log A_{rt-1} + \gamma \log IM_{rt-1} + \beta X_{rt-1} + \delta_1 HIV_{rt} + \delta_2 W_{rt} + s_r + \nu_t + \varepsilon_{rt} \tag{3.5}$$

$$\Delta \log IM_{rt} = \alpha(\Delta \log A_{rt-1})_{+} \gamma(\Delta \log IM_{rt-1}) + \beta(\Delta X_{rt-1}) + \delta_1 \Delta HIV_{rt} + \delta_2 \Delta W_{rt} + \Delta\nu_t + \Delta\varepsilon_{rt} \tag{3.6}$$

Equation (3.5) uses (two and three period) lagged differences of endogenous variables as instruments while (3.6) uses (once and twice) lagged levels. The system is preferred to the difference GMM because "exploiting the additional moment conditions in the level equations provides a dramatic improvement in the accuracy of estimates when the dependent variable is persistent" (Mishra & Newhouse, 2009). The authors note that the data suggests infant mortality is nearly a random walk, making the dependent variable persistent. It also makes any first-difference GMM estimator inappropriate since instruments will be weak. Furthermore, the authors also check to verify the number of instruments are appropriate using a Hansen's J test, verification of estimate coefficients using less instruments, and specification tests.

Results and Robustness Checks

Mishra and Newhouse (2009) begin with a scatter plot of infant mortality and health aid to demonstrate the weak, positive relationship. The authors suggest this is likely because of endogeneity, which is verified when comparing the subsequent regression. It also complements them with a visual.

Mishra and Newhouse (2009) begins with a simple OLS (without FE) of lagged health aid on infant mortality, finding a negative association with statistical significance. This is confirmed with a system GMM specification. However, the authors also note that both specifications are vulnerable to bias if differential time trends of health improvements are correlated with aid. This result is verified with a variety of robustness checks including AR1 and AR2 tests, the Hansen test, alternative samples, alternative instruments, alternative explanatory variables, removing outliers, additional controls, alternative lags, alternative dependent variables, and a FE model.

Mishra and Newhouse (2009) also conducts additional regressions using overall aid as the independent variable. Since aid may be fungible (depending on how tied they are to specific purposes), they may have identical outcomes to health aid. To this question, the GMM specification finds no significant effect even when disaggregating for type of broad type of aid (infrastructure, education, etc.). This excludes conflict aid, which is associated with an increase in infant mortality. This second result also is in line with previous studies, but the authors choose not the implement as many robustness checks. The given checks include Hansen test, AR2 test, comparing estimates to an OLS regression, and changing the lags.

An ancillary, final finding is that health aid specifically crowds in fiscal health spending while overall aid crowds out health spending. This is supported by separate regressions indicating such.

Analysis

Mishra and Newhouse's (2009) approach to estimating the effect of health aid on infant mortality is thoughtful and insightful. In the main result, the authors conduct an number of robustness checks, yet, in the other results, the authors take a more cursory approach. This study takes issue with their GMM regression results for the effect of overall aid on infant mortality, which they use to dismiss any claim that it can influence infant mortality. While not a bold claim, the authors make it using a GMM regression that utilizes a very small number of observations ($\tilde{1}00$); this may lead to a Type II error. Their final finding is similarly supported by a small number of observations in a simple OLS regression.

3.3.5 Synthesis

The foreign aid literature provides compelling evidence to answer questions about the effects of aid and its underlying mechanisms. Hansen and Tarp (2001) demonstrate that capital accumulation may be a key mechanism in the effect of foreign aid on growth; this supports theory that builds from the work of Chatterjee et al. (2003). Clemens et al. (2012) improves the methodology of Hansen and Tarp (2001) — along with similar papers — using a difference GMM. Alvi and Senbeta (2012) focus more on the potential relationship between foreign aid and poverty, suggesting growth and poverty-targeting qualities of projects as potential mechanisms. Additionally, they suggest that multilateral aid may be more effective and cast some doubt on the efficacy of loans. The latter claim is disputed. Mishra and Newhouse (2009) detail how health aid can reduce infant mortality by crowding in government health spending. A GMM regressions suggests overall aid may not be as effective in reducing infant mortality, but Mishra and Newhouse (2009) utilize a small number of observations or disaggregated by type, increasing the chance of Type II error. Contributions these articles, ideas regarding theory, and my own concerns, regarding methodological approach and theory will be utilized in this study's model specification.

3.4 Conclusion

The above review also supports the Theory chapter in suggesting that climate finance can influence poverty. Research on job creation by Wei et al. (2010) and Batini et al. (2022) reinforces the hypothesis in the Theory chapter that climate finance can create jobs. Wei et al. (2010) find that job creation results from renewable energy investments through energy efficiency gains, and Batini et al. (2022) estimates multipliers greater than one for renewable energy investments. It is then suggested that climate finance investments more broadly may create jobs given the parallels between the types of projects. The foreign aid literature also supports the importance of capital accumulation as a key mechanism in increasing growth and infant mortality — representative of health more broadly — is identified as another key vector (and potential proxy) of poverty (Hansen & Tarp, 2001; Mishra & Newhouse, 2009).

This chapter also closely discusses articles that utilize methods that would be also appropriate for this study. These articles primary utilize regression based approaches (Alvi & Senbeta, 2012; Bayramoglu et al., 2023; Clemens et al., 2012; Hansen & Tarp, 2001; Mishra & Newhouse, 2009). They typically begin with FE then IV. However, Clemens et al. (2012) broadly discusses concerns with instrument choices more broadly in the context of foreign aid. To mitigate such concerns, each article employs a varied set of robustness checks such as multiple specifications and statistical tests for instruments. Each article's approach to the subject is carefully considered in the context of this study's own methodology.

While these articles discuss closely related areas, they do not directly address nexus of climate finance and poverty. The most relevant articles, namely Bayramoglu et al. (2023) and Zhao et al. (2025), rely on Rio markers. Data from Fan et al. (2025) that builds from the keyword based approach of Michaelowa and Michaelowa (2011b) is utilized to address this concern and create a data source we can use to analyze climate finance. Additionally, I will evaluate the immediate impacts on other key vectors of poverty to understand underlying mechanisms unlike other articles.

Chapter 4

Data and Methods

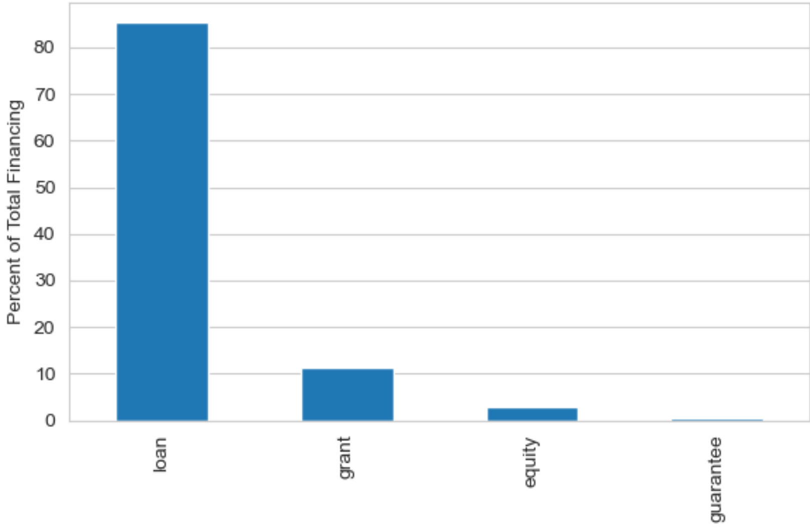
4.0.1 Data Description

I primarily rely on data by Fan et al. (2025) as a source of (multilateral) annual climate finance data. Before the release of this dataset, MDBs and MCFs publicly posted their project commitments on separate, independent data banks; this made it difficult to gather for analysis. Additionally, as further discussed by Michaelowa and Michaelowa (2011b), there are a number of concerns about the accuracy of the bilateral climate finance data from the OECD utilized by most existing studies. To correct for this, the authors use a machine learning model to compile this dataset together based on project descriptions (when possible). The author's findings roughly compare to other technical reports of total (multilateral) climate finance. Given this data was released in June 2025, there are any current academic articles utilizing it for analysis. Additionally, I argue that this dataset is representative of all international climate finance (or climate aid) considering the volume of multilateral aid relative to bilateral aid (Naran et al., 2025).

The percentage of loans relative to all financing instruments in real terms is 85.53%, and the majority of the remaining share, 11.27%, is disbursed via grants (see Fig 4.1).[1] Grants and equity have also become popular, particularly after 2020, and guaranties are small portion of overall lending (see Fig 4.2). Lastly, mitigation financing seems to be dominated by renewable energy projects with adaptation financing being dominated by disaster risk reduction and early warning system spending (see Fig. 4.5).

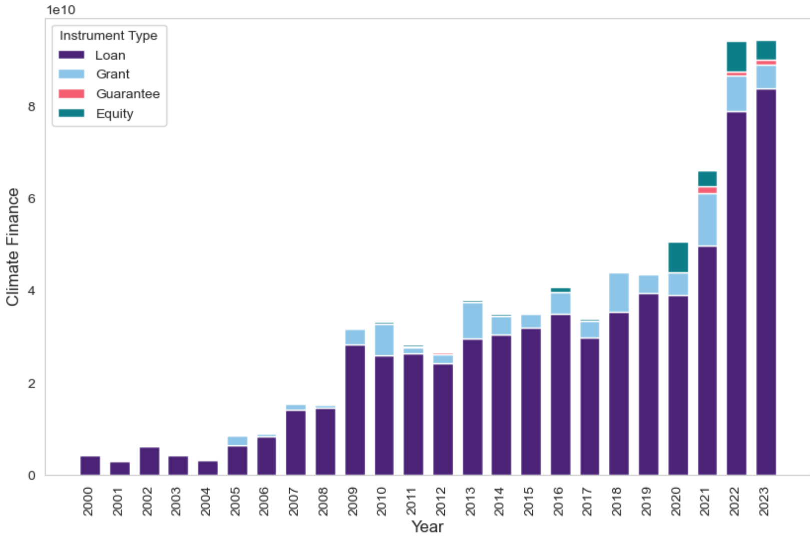
[1] Alvi and Senbeta (2012) find that most of the effects of climate finance on poverty are driven by grants and not loans. Given that following analysis utilizes aggregate commitments as the independent variable and comparing the relative size of loans versus other instruments, this will likely influence the results.

Figure 4.1: Relative Share of Financing Instruments (2000-2023)



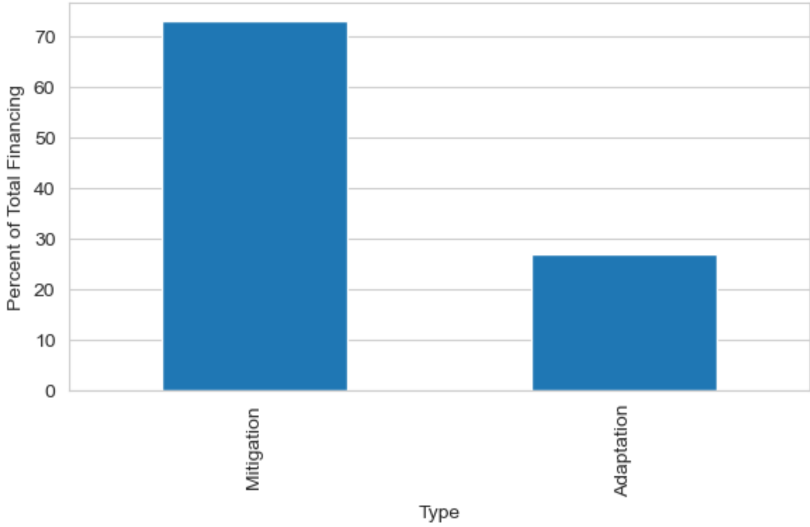
Note: Value of funds are in commitments and were adjusted to 2023 USD.

Figure 4.2: Climate Finance By Financial Instrument (2000–2023)



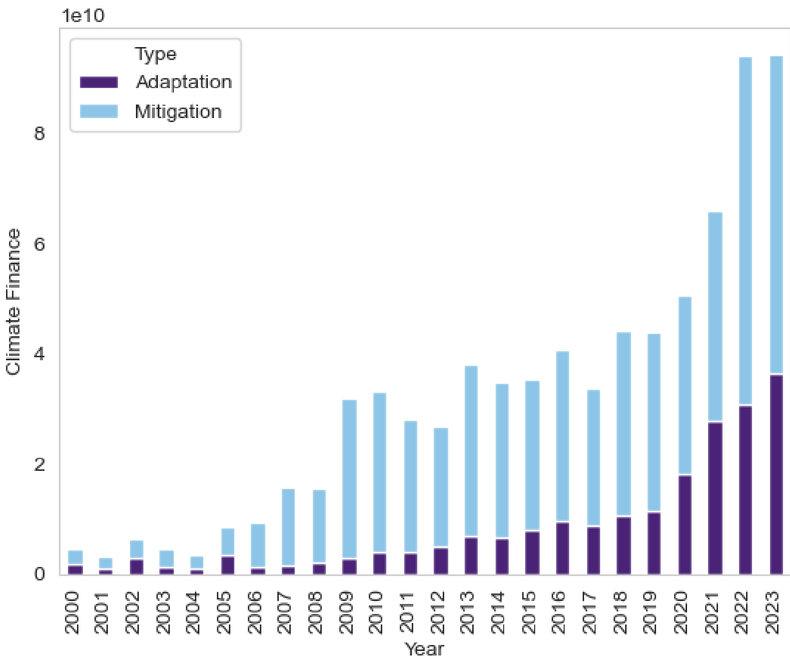
Note: Y-axis is scaled to be tens of billions of 2023 USD.

Figure 4.3: Relative Share of Financing by Purpose (2000–2023)



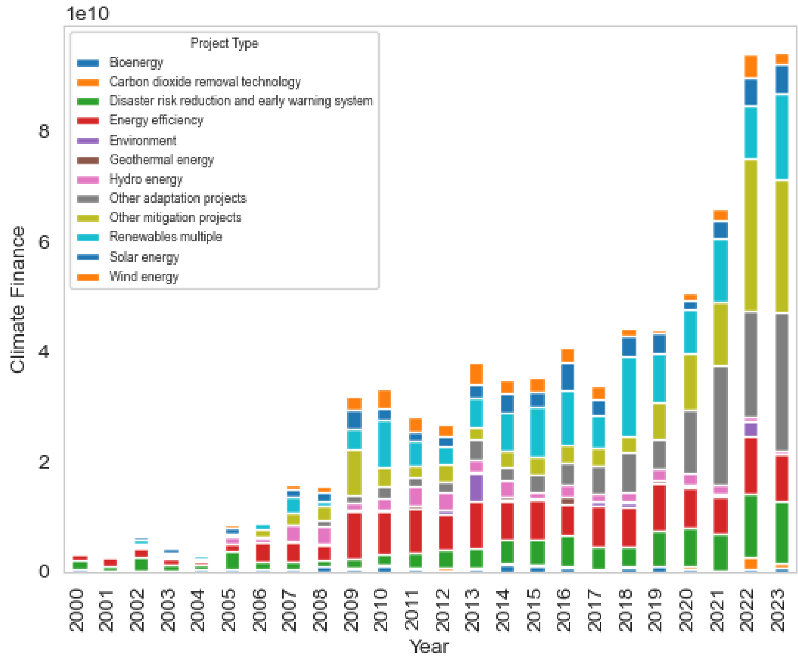
Note: Value of funds are in commitments and were adjusted to 2023 USD.

Figure 4.4: Multilateral Public Climate Finance (2000–2023)



Note: Y-axis is scaled to be tens of billions of 2023 USD.

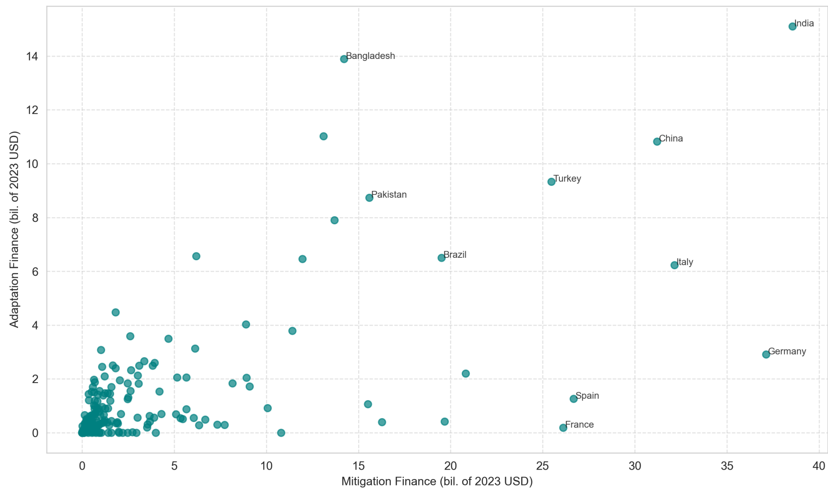
Figure 4.5: Climate Finance by Purpose (2000–2023)



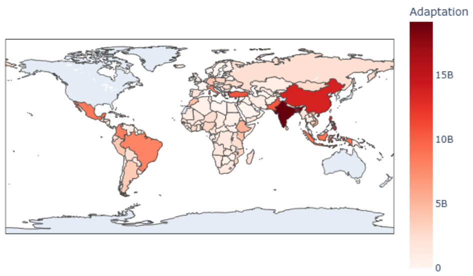
Note: Y-axis is scaled to be tens of billions of 2023 USD.

In recent years, the amount of adaptation spending is rising overall, but there is still a greater share devoted to mitigation over adaptation across all years (see Fig. 4.4 and 4.3). However, as a total aggregated over the 24-year sample period, most countries have received some combination of less than four billion (in 2023) USD for adaptation and/or mitigation (see Fig. 3). There are also a number of outliers in terms of the total volume of financing within the sample, particularly in countries with large populations. This includes India, China, Turkey, Bangladesh, Brazil, and others. By location, I observe that adaptation financing is concentrated China and India, and mitigation financing is concentrated in China, India, Europe, and Brazil. A list of all countries in the original sample is available in Table 4.1.

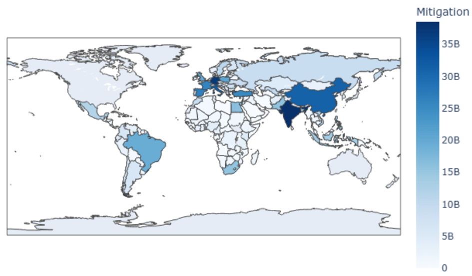
Figure 4.6: Adaptation and Mitigation Finance by Country (Total from 2000-2023)



(a) Adaption vs. Mitigation Financing



(b) Adaptation Financing by Country



(c) Mitigation Financing by Country

For brevity, definitions, transformations, and sources for the dependent variables and controls can be found in Table 4.2. The observations are on an annual basis, and the data sources include World Bank (2025), OECD (2025d), AIAS/OECD (2025), Fan et al. (2025), OCED (2026a), Center for Systemic Peace (2018), IMF (2025a), OECD (2025e), Muravyev (2014), OECD (2025a), WIPO (2025), International Labor Organization (2025), OECD (2025b), OECD (2026a), Notre Dame Global Adaptation Initiative (2025), OECD (2025c), IMF (2025b), Bank (n.d.), The Conference Board (2026), OECD (2026b), OCED (2026b), and The Heidelberg Institute for International Conflict (2023). Summary statistics for each can be found in Table 4.3.

The full dataset and a replication package is available here.[2]

[2] I opted not to use GitHub as there were limits on file size, and a number of files were downloaded locally.

Table 4.1: List of Countries

Afghanistan	Djibouti	Liberia	Saint Kitts and Nevis
Albania	Dominica	Libya	Saint Lucia
Algeria	Dominican Republic	Lithuania	Saint Vincent and the
Angola	Ecuador	Luxembourg	Grenadines
Antigua and Barbuda	Egypt	Macedonia	Samoa
Argentina	El Salvador	Madagascar	Sao Tome and Principe
Armenia	Equatorial Guinea	Malawi	Saudi Arabia
Austria	Eritrea	Malaysia	Senegal
Azerbaijan	Estonia	Maldives	Serbia
Bahamas	Eswatini	Mali	Seychelles
Bahrain	Ethiopia	Malta	Sierra Leone
Bangladesh	Fiji	Marshall Islands	Singapore
Barbados	Finland	Mauritania	Slovakia
Belarus	France	Mauritius	Slovenia
Belgium	Gabon	Mexico	Solomon Islands
Belize	Gambia	Micronesia, Federated States	Somalia
Benin	Georgia	of	South Africa
Bhutan	Germany	Moldova	South Sudan
Bolivia	Ghana	Mongolia	Spain
Bosnia and Herzegovina	Greece	Montenegro	Sri Lanka
Botswana	Grenada	Morocco	Sudan
Brazil	Guatemala	Mozambique	Suriname
Brunei	Guinea	Namibia	Sweden
Bulgaria	Guinea-Bissau	Nauru	Switzerland
Burkina Faso	Guyana	Nepal	Syria
Burma	Haiti	Netherlands	Tajikistan
Burundi	Honduras	Nicaragua	Tanzania
Cambodia	Hungary	Niger	Thailand
Cameroon	Iceland	Nigeria	Timor-Leste
Cape Verde	India	Niue	Togo
Central African Republic	Indonesia	North Korea	Tonga
Chad	Iran	Norway	Trinidad and Tobago
Chile	Iraq	Oman	Tunisia
China	Ireland	Pakistan	Turkey
Colombia	Israel	Palau	Turkmenistan
Comoros	Italy	Palestine	Tuvalu
Congo, Democratic Republic	Jamaica	Panama	Uganda
of the	Jordan	Papua New Guinea	Ukraine
Congo, Republic of the	Kazakhstan	Paraguay	United Kingdom
Cook Islands	Kenya	Peru	Uruguay
Costa Rica	Kiribati	Philippines	Uzbekistan
Cote d'Ivoire	Kuwait	Poland	Vanuatu
Croatia	Kyrgyzstan	Portugal	Venezuela
Cuba	Laos	Qatar	Vietnam
Cyprus	Latvia	Romania	Yemen
Czech Republic	Lebanon	Russia	Zambia
Denmark	Lesotho	Rwanda	Zimbabwe

Table 4.2: Data Description and Sources*

Variable Name	Description	Source
Aid	Official development assistance (ODA)	World Bank (2025)
ALMP	Active labor market program spending in current USD.[k]	OECD (2025d)
Age Dependency Ratio	% of working age population	World Bank (2025)
Bargaining Index	Centralization of bargaining	AIAS/OECD (2025)
Child Spending	Early education and care in current USD	OECD (2025d)
Climate Finance	Total volume in mil. of (current) USD by financial instrument	Fan et al. (2025)
Coverage by Bargaining Agreements	% of total employed population	OCED (2026a)
Coordination	Coordination and government intervention in wage bargaining	AIAS/OECD (2025)
Democracy Index	DEMOC, Polity IV	Center for Systemic Peace (2018)
Disaggregated Capital Formation	Government, Private, Public-private partnership	IMF (2025a)
Domestic Credit to Private Sector	Current USD	World Bank (2025)
Employment Protection	Consolidated index of strictness of employment legislation.[g]	OECD (2025e) and Muravyev (2014)
FDI	Net foreign direct investment in current USD	World Bank (2025)
Fertility Rate	Total number of children a woman would bear	World Bank (2025)
Full-time Employment	Persons	OECD (2026b)
GDP	Gross domestic product at current USD	World Bank (2025)
Gini	Gini coefficient	World Bank (2025)
Health Aid	Current USD disbursements.[l]	OECD (2025a)
HIV	Incidence of HIV (per 1,000 uninfected population)	World Bank (2025)
Incapacity spending	Current USD	OECD (2025d)
Infant Mortality	Rate per 1000 live births	World Bank (2025)
Inflation	CPI, annual % change	World Bank (2025)
Innovation	Global Innovation Index.[a]	WIPO (2025)
Labor force participation rate	Estimates Nov. 2024, % of sub-population.[b]	International Labor Organization (2025)
Life Expectancy	Mean life expectancy	World Bank (2025)
LTI	Real long term interest rate (nominal LTI minus inflation)	OECD (2025b)
Maternity Leave	Total length in months (Excluded).[c]	OECD (2026a)
ND-GAIN	Notre Dame Gain Index	Notre Dame Global Adaptation Initiative (2025)
Net Trade	Value of exports minus imports (current USD).[d]	World Bank (2025)
Official flows	Concessional and non-concessional aid	OECD (2025c)
Output Gap	% of potential GDP	IMF (2025b)
Poverty	Headcount, Gap, Squared Gap at $2.15 (2021 PPP USD)	Bank (n.d.)
Schooling	Total upper secondary and tertiary employment.[e]	World Bank (2025)
Tax Wedge	Ratio of taxes paid to total labor cost for the employer.[f]	OECD (2025b)
Tertiary/Service Employment	Ratio of employed people (relative share).[h]	World Bank (2025)
TFP Growth	Annualized change in the natural log of total factor productivity	The Conference Board (2026)
Unemployment Rate	% of subpopulation, 15+	OECD (2026b)
Unemployment Benefits	Net replacement rate.[i, j]	OCED (2026b)
Urbanization	% of population	World Bank (2025)
War	Dummy variable representing war	The Heidelberg Institute for International Conflict (2023)

*All sources using current USD converted to constant USD using US CPI data from World Bank (2025). Variables used in poverty regressions were converted to constant 2021 USD. [a] Substituted for the original authors' measure of relative price of investment and routinization due to survey-level data constraints.

[b] Measure disaggregated to workers ages 15+, 15–24, and 55–64. Prime age (25–54) is further disaggregated by sex. I utilize workers ages 55–64 rather than 55+ based on data availability.

[c] Excluded from final regressions due to insufficient coverage in current OECD datasets.

[d] Sourced from World Development Indicators (WDI) rather than IMF World Economic Outlook.

[e] Substituted for Barro-Lee Educational Attainment dataset due to coverage constraints over the time horizon.

[f] Calculated at 67% of average earnings as 100% is no longer provided by the OECD.

[g] Average of synthetic indicators of employment protection for regular and temporary contracts from the OECD (2025e) excluding Estonia from 2000-2007, which is from Muravyev (2014). The latter data source is available upon request on from the International Data Service Center.

[h] Calculated by multiplying WDI share of total employment by OECD total persons employed, then dividing by total employment in each sector.

[i] For LFPR regressions: Average of 67% and 100% average wage across multiple family types, three durations of unemployment (1, 6, and 12 months), and spouse assumes minimum wage earnings.

[j] For unemployment regressions: Average net benefits (excluding social assistance) for 100% of average earnings, two family situations, and two months of unemployment duration following Perone (2022).

[k] For unemployment regressions, expenditure is restricted to public employment services, training, and employment incentives. No restrictions applied for labor force participation regressions.

[l] Restricted health aid to CRS codes 12110, 12181, 12182, 12191, 12220, 12230, 12240, 12250, 12261, and 12281 following Mishra and Newhouse (2009).

Table 4.3: Summary Statistics

	count	mean	sd	min	max
Country Name	0	.	.	.	.
Year	1037	2012.426	6.403848	2000	2023
Total Climate Finance	1037	3.71e+08	6.71e+08	0	5.46e+09
Loan Amount	1037	3.50e+08	6.52e+08	-1.86e+07	5.20e+09
Grant Amount	1037	1.58e+07	5.03e+07	0	6.71e+08
Guarantee Amount	1037	812282.6	8242378	0	1.44e+08
Equity Amount	1037	4654477	2.48e+07	0	2.73e+08
Country Code	0	.	.	.	.
Age dependency ratio	1037	52.65243	7.257276	17.45307	90.76306
Domestic Credit to Private Sector	970	1.56e+19	2.18e+20	3.73e+08	4.72e+21
Net Foreign Direct Investment Inflows	1027	1.37e+09	3.84e+10	-3.71e+11	2.36e+11
Gross Domestic Product	1037	6.31e+11	1.04e+12	1.74e+09	5.39e+12
Gini Index	1037	37.08284	8.919719	23.2	61.6
Gross Fixed Capital Formation	1036	1.36e+11	2.14e+11	4.76e+08	1.13e+12
HIV Prevalence	782	.1603453	.1191692	.01	1.4
Inflation Rate (CPI)	1031	3.556655	5.10982	-4.447547	96.09638
Life Expectancy at Birth	1037	77.21549	4.334889	58.591	83.90488
Infant Mortality Rate	1037	8.520444	8.604016	1.6	68.5
Net Trade	1029	.0067409	.0993041	-.3533525	.437729
Total Population	1037	2.31e+07	3.71e+07	89949	2.11e+08
Government Effectiveness Index	1018	.6998041	.8635606	-1.659284	2.347191
Political Stability Index	1018	.3824412	.7343037	-2.376027	1.720112
Regulatory Quality Index	1018	.7855207	.754096	-1.259179	2.040489
War/Conflict Indicator	1037	.0086789	.0928001	0	1
Mean Years of Schooling	677	10.20021	2.289817	3.65	14.26
ND-GAIN	1037	56.80502	10.38572	33.01009	78.31512
Health Aid Disbursements	337	1.99e+07	3.56e+07	72408.45	4.18e+08
Fertility Rate	1037	1.857392	.5595513	1.08	4.583
Secondary Education	952	59.95119	21.19613	10.59	96.84
Teritary Education	883	23.2702	10.10626	3.41	64.09
ODA	371	4.80e+08	4.91e+08	-4.27e+08	3.93e+09
Official Flows	362	1.08e+09	1.73e+09	-1.13e+10	8.90e+09
Urbanization	1037	72.38648	12.20218	20.02043	99.03101
LFPR (25-54 yrs, Female)	1035	74.67346	10.92755	40.83	90.513
LFPR (25-54 yrs, Male)	1035	91.92897	2.842221	78.6	99.53
LFPR (55-64 yrs, All)	1035	57.44874	12.84032	24.892	87.555
LFPR (15-24 yrs, All)	1035	45.37967	13.29563	19.652	83.028
LFPR (15+ yrs, All)	1035	60.88135	6.059651	47.207	87.498
Democracy Index	368	8.429348	5.289484	-88	10
Output Gap	452	-.4692699	3.57991	-18.64	15.085
Bargaining Index	619	2.102989	.9906496	.875	4.875
Global Innovation Index	625	42.91429	11.73254	-1	68.4
Incapacity Spending	674	2.519602	1.483983	0	7.145
Old Age Spending	674	8.03299	3.367671	0	15.43
Part Time Employment	761	13.14438	7.269284	1.132	38.437
Unemployment Benefits	623	71.23016	9.79743	49.35185	92.5
Maternity Leave	340	57.08088	46.08811	12	166
Early Childhood Education and Care Spending	671	.6196051	.3867198	0	1.831
ALMP	629	4.18e-11	1.64e-10	0	1.41e-09
Average Tax Wedge	644	19.6959	11.75345	-17.23415	39.91134
Total Employment	761	11266.13	18280.4	146.227	100573.5
Government Captial Formation	856	3.85184	1.993116	1.294167	22.92551
PPP Captial Formation	721	.2714523	.4592649	0	3.511587
Private Captial Formation	856	16.95603	4.403908	4.165482	45.15393
Poverty Headcount	1037	.0223436	.0449743	0	.6323893
Poverty Gap	1037	.0090827	.0206295	0	.3355999
Squared Poverty Gap	1037	.0053343	.0129675	0	.2181244
Observations	1037				

4.0.2 Empirical Strategy

In the following section, I will estimate the effects of climate finance on capital formation, infant mortality, labor force participation, output, and poverty, utilizing TWFE with clustered standard errors across all specifications. These are collected at the country level on an annual basis from a wide variety of sources. If a variable contains monetary units, there were adjusted from current to 2023 USD as it is the most recent available year. This was transformed using CPI data for the USA from World Bank (2025). However, for the poverty regression I opt for 2021 USD given the most recent poverty line data was from 2021.

For each dependent variable, I choose to replicate other articles with the same predictors, and I make adjustments to incorporate climate finance as a key predictor. I also attempt to use the same control variables from each article faithfully using updated, modern datasets from the same sources if possible. Throughout each regression, I also attempt to correct for multi-colinearity by monitoring each regression's variance inflation factor (VIF). If the VIF is above 10 and there are some variables that are highly correlated, I remove the theoretically less important variable to reduce the VIF. Additionally, all regressions use clustered standard errors to account for heterogeneity and serial correlation, and I test for the importance of a country's climate vulnerability/adaptation to climate change using the Notre Dame Global Adaptation Initiative Index (ND-GAIN).

For each possible development impact of climate finance, I begin by estimating by estimating OLS regressions with year, t, and country, i, fixed effects. I choose to have some regressions without country fixed effects because of results from the Hausman test. I begin with standard OLS regressions with fixed effects following Hansen and Tarp (2001). The preferred specification for sets of regressions without lags can be expressed as

$$Y_{it} = \beta_0 + \beta_1 CF_{it} + \beta' X_{it} + \gamma_t + \varepsilon_{it} \tag{4.1}$$

where Y_{it} is the log or inverse hyperbolic sine (IHS) transformed dependent variable to account for skewness. I prefer log transformations excluding the necessary case when zeros are present, in which case I use a IHS transformation. Furthermore, β_0 is the constant, CF_{it} is aggregated IHS-transformed climate finance, X is the vector of control variables, λ_t is country and/or time fixed effects, and ε_{it} is the error term. By aggregate, I am referring to the aggregate value of climate finance commitments across the measured categories in the dataset: loans, grants, equities, and grants.

To evaluate the importance of time horizons on potential impacts, I isolate short run effects using (the flow of) real climate finance commitments for a given year and long run effects with (the stock of) real accumulated climate finance commitments across the observed time period (2000-2023). I also lag climate finance one year, five years, and ten years; given that project commitments are generally disbursed within two years, I expect the one-year lag to have the strongest impact (Avellán, Calderon, & Lotti, 2021). This improves upon the "two-period smoothing approach" of other papers that attempt to reconstruct disbursements from commitments (Lee et al., 2022).

For output, I estimate the impact of climate finance commitments (for a given year) on GDP per capita growth following using equation (4.2) and follwing Hansen and Tarp (2001).[3] All specifications for each regression uses a common set of controls: inflation, net trade, the lagged growth rate, and log initial GDP per capita.[4] I also control for the policy environment (WGI political stability index), foreign aid, private flows, capital formation, and human capital (mean years of schooling). As GDP per capita growth appears normally distributed, I do not apply any transformations.

Similarly to output, I also test impacts on three differment measures of poverty – headcount, gap, and squared poverty gap – by using only IHS transformed climate finance commitments (2021 USD) using equation (4.2).[5] The poverty line for the following statistics is set at $2.15 a day in 2021 PPP USD. For this set of regressions only, I opt to use 2021 USD for interoperability.

Given that health effects may occur over a long time horizon, I then estimate the impact of (millions of 2023 USD of) climate finance on infant mortality using fixed effects with lags following Mishra and Newhouse (2009).[6] The regression equation is specified as

$$I_{it} = \beta_0 + \beta_1 CFPC_{it} + \beta_2 H_{it-1} + \zeta' A_{it-1} + \theta' B_{it} + \alpha_i + \gamma_t + \varepsilon_{it} \tag{4.2}$$

such that I_{it} is log infant mortality (as a proxy for health with the transformation following the original authors), H_{it-1} is log health aid per captia lagged t years, A_{it-1} is the vector of lagged and log transformed

[3] I use a combination of controls from Table 5.3 and Table 5.4 given more recent data availability.

[4] Log initial GDP per capita is only present when fixed effects are not employed

[5] Data availability limited the number of countries available in the estimation sample to 17. They are (in ISO-3 codes for brevity) ALB, BOL, BRA, CHL, COL, DOM, ECU, EST, HUN, MEX, NIC, PAN, PER, POL, SLV, TUR, and VEN.

[6] Specifically, I attempt to replicate and amend the OLS regression found in Table 5.1 of their paper.

control variables, B_{it} is the vector of control variables that are not lagged, α_i is country fixed effects, γ_i is time fixed effects, and ε_{it} is the error term. $CFPC_{it}$ is IHS transformed climate finance per capita following Mishra and Newhouse's (2009) treatment of health aid. I also control for lagged log health aid per capita, lagged log infant mortality, lagged log GDP per capita, lagged log population, lagged log fertility rate, a dummy variable for war in that year, and HIV incidence.

Following Perone (2022), I estimate the impact of climate finance on the log unemployment rate of all workers 15 years or older (All, 15+) using equation (4.2). I log the unemployment rate because of the skewness in the distribution, which contrasts from the original paper. Control variables include the ratio of gross capital formation to GDP, employment protection, coverage of workers by bargaining agreements, coordination of bargaining, the replacement rate of unemployment benefits, active labor market programs (ALMP), the real long term interest rate, TFP growth, and terms of trade.

Following Grigoli, Koczan, and Topalova (2018), I estimate the relationship between climate finance (mil. 2023 USD) and different subsets of labor force using equation (4.2).[7] For all disaggregated regressions (Tables 5.8, 5.9, and 5.10), I include the output gap, innovation (Global Innovation Index), net trade as as share of GDP, urbanization, completion of upper secondary education (% of the population), completion of tertiary secondary education (% of the population), the average tax wedge, unemployment benefits, spending on active labor market programs, unionization, and the centralization of bargaining. Specification (2) for each also includes old age and incapacity spending as additional controls. Specification (5) includes statutory maternity leave, public spending on early childhood education and care, and the share of part-time employment. Labor force participation is expressed as the log participation rate of the population in the respective subcategory given the skewness of the distributions of some of the dependent variables. It is important to note that data in the estimation sample for the labor force participation regressions only includes OECD countries because of data limitations.[8]

Following Hansen and Tarp (2001), I examine the impact of aggregate climate finance commitments on (the flow of) gross fixed capital formation (see Table 5.13).[9] Control variables include official flows, private flows, the ND-GAIN index, inflation, net trade, private credit, population, and the WGI political

[7] I attempt to replicate Table 5.1 from Grigoli et al. (2018) and amend it to include climate finance. Given data availability, I some adjustments that I note in Table 4.2

[8] I obtained data for the following countries (represented by ISO-3 codes): AUT, BEL, DEU, DNK, ESP, EST, FIN, FRA, GBR, GRC, IRL, ITA, LUX, NLD, NOR, PRT, SVK, SVN, and SWE.

[9] Gross fixed capital formation is defined as,"gross capital formation includes acquisitions less disposals of produced assets for purposes of fixed capital formation, inventories or valuables"(World Bank, 2025) For the disaggregated analysis, the data are sourced separetely but defined similarly IMF (2025a)

stability index.[10] The Hausmen test also indicated country fixed effects were unnecessary, and I also conduct an additional heterogeneity analysis on the components of gross capital formation.

[10]Official flows are the sum of official development assistance and other official flows. It can also be interpreted as as concessional foreign aid and other bilateral/multilateral aid that is non-concessional in nature. The ND-GAIN index is a measure of a country's vulnerability to climate and ability to adapt (Notre Dame Global Adaptation Initiative, 2025). These two covariates were introduced to account for donor and investor preferences.

Chapter 5

Results

5.0.1 Output

I first estimate short-run effects in specifications (1), (2), and (3) (see Table 5.1). I begin with a no control specification (1), add controls (2), and then add year fixed effects (3). I find no statistically significant effects excluding (1). To estimate the long-run effects, I then accumulate climate finance (4), and try one-year (5), five-year (6), and ten-year lags (7) on annual climate finance commitments. In the preferred specification (5), I find a that 1% increase in climate finance commitments leads to a 0.244% decrease in the GDP per capita growth rate at the 10% level of significance. This is a sizable and economically meaningful decrease in GDP per capita. Specifications (4), (6), and (7) are not statistically significant. I also estimate an interaction with the ND-GAIN index as a measure of institutional quality in response to climate change vulnerability (i.e. adaptation readiness). I find no statistically significant results, and these results can be found in Appendix A.

5.0.2 Poverty

I find that there is a small, statistically significant relationship between IHS transformed climate finance commitments and poverty (specification (1), (2), and (3)) (see Table 5.2). However, this becomes insignificant upon the introduction of year and country fixed effects. I then use cumulative financing to estimate long run effects (see Table 5.3).[1] The preferred specifications – (2), (4), and (6) – have coeffi-

[1] It is important to note that this makes climate finance a stock variable, which is being compared the poverty rate, a flow variable.

Table 5.1: IHS Climate Finance and GDP per Capita Growth

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
IHS Climate Finance	-0.121***	-0.074	0.016				
	0.026	0.143	0.137				
IHS Cumulative Climate Finance				-0.430			
				0.296			
L.IHS Climate Finance					-0.224*		
					0.125		
L5.IHS Climate Finance						-0.031	
						0.089	
L10.IHS Climate Finance							0.015
							0.050
adj. R^2	0.006	0.143	0.513	0.514	0.519	0.551	0.559
Controls		✓	✓	✓	✓	✓	✓
FE			✓	✓	✓	✓	✓
Observations	4149	357	357	357	357	318	236

Climate finance is in mil. 2021 USD and is IHS transformed. The L refers to the number of lags. FE refers to country and year fixed effects. Standard errors clustered at the country level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

cients that are not statistically significant. The results for climate finance lagged one year are remarkably similar in both coefficients and statistical significance (see Table 5.4).

The key difference between the short run results (Table 5.2) and time varying results (Table 5.3 and Table 5.4) only lies in the significance of the coefficient of specification (5), which is the squared poverty gap without fixed effects. I also estimate the effect on poverty interacted with ND-GAIN, and I find some statistically significant results similar to the above. Given that the coefficients are very small and not economically meaningful, I include these regressions in Appendix A.

Table 5.2: Short Run Effect of Climate Finance on Poverty

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable	Poverty Rate		Poverty Gap		Squared Gap	
IHS Climate Finance	-0.002***	-0.000	-0.001***	-0.000	-0.001***	-0.000
	0.000	0.001	0.000	0.000	0.000	0.000
Controls	✓	✓	✓	✓	✓	✓
FE		✓		✓		✓
Observations	163	163	163	163	163	163

Climate finance is in mil. 2021 USD and is IHS transformed. FE refers to country and year fixed effects. Standard errors clustered at the country level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

,

Table 5.3: Long Run Effect of Climate Finance on Poverty

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable	Poverty Rate		Poverty Gap		Squared Gap	
IHS Cumulative Climate Finance	-0.002*	0.001	-0.001*	0.001	-0.001	0.001
	0.001	0.001	0.000	0.001	0.000	0.000
Controls	✓	✓	✓	✓	✓	✓
FE		✓		✓		✓
Observations	163	163	163	163	163	163

Climate finance is in mil. 2021 USD and is IHS transformed. FE refers to country and year fixed effects. Standard errors clustered at the country level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table 5.4: One-Year Lagged Effect of Climate Finance on Poverty

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable	Headcount		Poverty Gap		Squared Gap	
L.IHS Climate Finance	-0.002***	-0.000	-0.001**	-0.000	-0.001*	-0.000
	0.001	0.001	0.000	0.000	0.000	0.000
Controls	✓	✓	✓	✓	✓	✓
FE		✓		✓		✓
Observations	130	130	130	130	130	130

Climate finance is in mil. 2021 USD and is IHS transformed. FE refers to country and year fixed effects. Standard errors clustered at the country level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

5.0.3 Infant Mortality

In short, excluding the no control specification, I find that no statistically significant results across time various time horizons (see Table 5.5). The high adjusted R^2 from specifications (2) to (7) highlights the high explanatory power of the regression. I also run an interaction with ND-GAIN, but I find no statistically significant results across all specifications. I include this in Appendix A.

Table 5.5: Climate Finance per Capita (CFPC) and Log Infant Mortality

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
IHS CFPC	-0.107***	0.000	-0.000				
	0.018	0.001	0.000				
IHS CFPC				-0.001			
				0.001			
L.IHS CFPC					-0.000		
					0.001		
L5.IHS CFPC						-0.000	
						0.001	
L10.IHS CFPC							0.001
							0.001
Controls		✓	✓	✓	✓	✓	✓
Year FE			✓	✓	✓	✓	✓
Adj. R^2	0.047	0.993	0.933	0.933	0.933	0.913	0.833
Observations	4392	2314	2314	2314	2314	2090	1522

Climate finance is in mil. 2021 USD and is IHS transformed. Year FE refers to year fixed effects. The L indicates the number of lagged years. Standard errors clustered at the country level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

5.0.4 Unemployment

I find statistically results for climate finance committed in the same year in specification (3) (see Table 5.6). To estimate long run effects, I accumulate climate finance in specification (4), and I find a no statistically significant results. I then lag climate finance one year in specification (5), and I find that a 1% increase in climate finance leads to a 0.007% increase in the unemployment in the subsequent year at the 10% level of significance. In five years (6), I find a 1% increase in climate finance leads to a 0.004% increase in the unemployment in five years at the 5% level of significance. In 10 years there is no statistically significant effect.

In Table 5.7, I estimate the interaction between climate finance and ND-GAIN. In specification (1), I find the coefficient on ND-GAIN to be significant. This suggests that the effect of a country's institutional response and vulnerability to climate change on the unemployment rate in the same year is significantly impacted by its access to climate finance. In specifications (2) to (4), I test for long run impacts through accumulated effects (2), a one-year lag (3), and a five-year lag (4). Each specification finds a positive marginal effect of climate finance dependent on the value of ND-GAIN, with the largest effect observed in the subsequent year.

Table 5.6: IHS Climate Finance (CF) and Log Unemployment Rate

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
IHS CF	0.011***	0.007**	0.005				
	0.002	0.003	0.004				
IHS Cumumulative CF				0.006			
				0.009			
L.IHS CF					0.007*		
					0.004		
L5.IHS CF						0.004**	
						0.002	
L10.IHS CF							0.000
							0.002
Controls		✓	✓	✓	✓	✓	✓
FE			✓	✓	✓	✓	✓
Adj. R^2	0.036	0.437					
Overall R^2			0.509	0.503	0.514	0.495	0.612
Observations	914	259	259	259	259	218	150

Climate finance is in mil. 2021 USD and is IHS transformed. The L indicates the number of lagged years. All,15+ indicates all workers regardless of gender 15 years or older. FE are year fixed effects. Standard errors clustered at the country level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table 5.7: Climate Finance, Adaptation Readiness (ND-GAIN), and Unemployment

	(1)	(2)	(3)	(4)
IHS Climate Finance	-0.014			
	0.027			
ND-GAIN	-0.027**	-0.059***		
	0.013	0.020		
IHS Climate Finance × ND-GAIN	0.000			
	0.000			
IHS Cumulative Finance		-0.104**		
		0.051		
IHS Cumulative Finance × ND-GAIN		0.002**		
		0.001		
L.ND-GAIN			-0.039***	
			0.014	
L.IHS Climate Finance			-0.056**	
			0.026	
L5.IHS Climate Finance				-0.040*
				0.021
L.IHS Climate Finance × L.ND-GAIN			0.001**	
			0.000	
L5.ND-GAIN				-0.030***
				0.009
L5.IHS Climate Finance × L5.ND-GAIN				0.001**
				0.000
Controls	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Overall R^2	0.563	0.577	0.584	0.669
Observations	259	259	259	218

Climate finance is in mil. 2021 USD and is IHS transformed. The L indicates the number of lagged years. All,15+ indicates all workers regardless of gender 15 years or older. FE are year fixed effects. Standard errors clustered at the country level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

5.0.5 Labor Force Participation Rate

I examine short run effects by using IHS transformed commitments for a given year (see Table 5.8). For the workers 15 years or older and workers between 25-54, there is no evidence of an effect. However, this differs for prime age workers. For men, there is a small but statistically significant effect, and, for prime age women, the effect size is at least twice. Additionally, for workers 55-64, a 1% increase in climate finance leads to a 0.003% in the labor force participation rate. In general, these are modest effects on the participation rate for each subcategory. In Table 5.9, I utilize a one-year lag, and I find the effects are generally similar.

Table 5.8: Short Run Effects of Climate Finance on Log Labor Force Participation

	(1)	(2)	(3)	(4)	(5)
Dependent Variable	All 15+	All 55-64	All 15-24	Male 25-54	Female 25-54
IHS Climate Finance	-0.001	0.003*	0.000	-0.000*	-0.001**
	0.001	0.001	0.002	0.000	0.001
Controls	✓	✓	✓	✓	✓
Observations	247	247	247	247	151

Standard errors clustered at the country level. *, **, *** denote significance at the 10, 5, and 1 percent levels

Table 5.9: One-Year Lagged Climate Finance and Log Labor Force Participation

	(1)	(2)	(3)	(4)	(5)
Dependent Variable	All 15+	All 55-64	All 15-24	Male 25-54	Female 25-54
L.IHS Climate Finance	-0.000	0.002*	0.001	-0.000*	-0.001*
	0.001	0.001	0.002	0.000	0.001
Controls	✓	✓	✓	✓	✓
Observations	247	247	247	247	151

All specifications have covariates as described. The L indicates the lag is of one year. Standard errors clustered at the country level. *, **, *** denote significance at the 10, 5, and 1 percent levels

I then accumulate climate finance to to estimate the long run effects (see Table 5.10). Specifications (1), (2), (3), and (4) find no statistically significant effects. I specification (5), I find that a 1% increase in climate finance leads to a 0.003% decrease in labor force participation for prime age women. This effect size is much larger than specification (5) in Table 5.7. Additionally, the direction of the coefficients across the regression of Tables 5.8, 5.9, and 5.10 are generally inconsistent with the hypothesis.

Table 5.10: Cumulative Climate Finance (IHS) and Log Labor Force Participation

	(1)	(2)	(3)	(4)	(5)
Dependent Variable	All 15+	All 55-64	All 15-24	Male 25-54	Female 25-54
IHS Cumulative Climate Finance	-0.001	0.002	0.002	0.000	-0.003*
	0.002	0.004	0.005	0.000	0.002
Controls	✓	✓	✓	✓	✓
Observations	247	247	247	247	151

All specifications have covariates as described. Standard errors clustered at the country level. *, **, *** denote significance at the 10, 5, and 1 percent levels

To estimate the impact of climate finance on the overall labor force over a longer time horizon, I also introduce a larger lag structure in Table 5.11. I use the same controls as the previous labor force regressions. The regression coefficients are not statistically significant across all horizons. These include the climate finance committed the same year (3), accumulated climate finance (4), lagged one year (5), lagged five years (6), and lagged 10 years (7). The coefficients are insignificant in all specifications.

In Table 5.13, I utilize the same lag structure with an interaction term. I find the the coefficient on ND-GAIN to be significant at the 1% level across all specifications. This suggests that the effect of a country's institutional response and vulnerability to climate change on the labor force participation rate in the same year is significantly impacted by its access to climate finance.

Table 5.11: Climate Finance and Log Labor Force Participation Rates (15+)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
IHS CF	0.001	-0.001	-0.001				
	0.001	0.001	0.001				
IHS Cumulative CF				-0.001			
				0.002			
L.IHS CF					-0.000		
					0.001		
L5.IHS CF						-0.001	
						0.001	
L10.IHS CF							-0.001
							0.001
Controls		✓	✓	✓	✓	✓	✓
Year FE				✓	✓	✓	✓
Adj. R^2	0.000	0.590					
Overall R^2			0.659	0.660	0.656	0.686	0.712
Observations	4124	247	247	247	247	210	141

Climate finance is in mil. 2021 USD and is IHS transformed. The L indicates the number of lagged years. All,15+ indicates all workers regardless of gender 15 years or older. FE are year fixed effects. Standard errors clustered at the country level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

5.0.6 Capital Formation

In Table 5.13, Specification (3) finds that a 1% increase in cumulative climate finance corresponds to a 0.065% increase in capital formation at the 5% level of significance. While this pattern is consistent with the hypothesis that climate-related investments may support capital accumulation, the magnitude of the estimated coefficients suggests that the effect is economically modest. In Table 5.14, specification (3) finds a much smaller but significant effect of climate finance on gross capital formation in the subsequent year.

Table 5.12: Climate Finance (CF), Adaptation Readiness (ND-GAIN), and Log Labor Force Participation Rates (15+)

	(1)	(2)	(3)	(4)
IHS CF	-0.008			
	0.006			
gain	0.009***	0.011***		
	0.002	0.002		
IHS CF × gain	0.000			
	0.000			
IHS Cumulative CF		0.001		
		0.006		
IHS Cumulative CF × gain		-0.000		
		0.000		
L.gain			0.010***	
			0.002	
L.IHS CF			-0.003	
			0.005	
L5.IHS CF				-0.005
				0.005
L.IHS CF × L.gain			0.000	
			0.000	
L5.gain				0.009***
				0.002
L5.IHS CF × L5.gain				0.000
				0.000
Controls	✓	✓	✓	✓
Year FE		✓	✓	✓
Overall R^2	0.762	0.757	0.758	0.786
Observations	247.000	247.000	247.000	210.000

Climate finance is in mil. 2021 USD and is IHS transformed. The L indicates the number of lagged years. All,15+ indicates all workers regardless of gender 15 years or older. FE are year fixed effects. Standard errors clustered at the country level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table 5.13: Climate Finance and Log Gross Capital Formation

	(1)	(2)	(3)
IHS Climate Finance	0.046***	0.047***	0.005
	0.011	0.012	0.004
adj. R^2	0.033	0.694	0.644
IHS Cumulative Climate Finance	0.082***	0.217***	0.065**
	0.015	0.033	0.032
adj. R^2	0.362	0.736	0.658
Controls		✓	✓
FE			✓
Observations	3614.000	586.000	586.000

Climate finance is in mil. 2023 USD and capital formation in 2023 USD. No scale factor was applied variables that were IHS transformed. All regressions include country and year fixed effects (FE). Standard errors clustered at the country level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Table 5.14: Climate Finance and Log Gross Capital Formation

	(1)	(2)	(3)
L.IHS Climate Finance	0.045^{***}	0.037^{***}	0.006^{*}
	0.012	0.011	0.003
adj. R^2	0.034	0.688	0.645
Controls		✓	✓
FE			✓
Observations	3501.000	586.000	586.000

Climate finance is in mil. 2023 USD and capital formation in 2023 USD. No scale factor was applied variables that were IHS transformed. All regressions include country and year fixed effects (FE). Standard errors clustered at the country level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

Considering that gross capital formation is an aggregation between governments and the private sector, I conduct an additional heterogeneity analysis in Table 5.15.[2] Once again, the effects of climate finance are modest economically across the results. Specification (2) finds that a 1% increase in climate finance is associated with a 0.006% increase in private capital formation. I find no statistically effects for cumulative climate finance or climate finance lagged one year.

Table 5.15: Capital Formation in the Public Sector, Private Sector, and Public-Private Partnerships (PPPs)

	(1) Public (Log)	(2) Private (Log)	(3) PPPs (IHS)
IHS Climate Finance	-0.005	0.006**	-0.002
	0.004	0.003	0.003
adj. R^2	0.074	0.245	0.059
IHS Cumulative Climate Finance	-0.004	0.000	0.005
	0.016	0.011	0.012
adj. R^2	0.068	0.231	0.059
L.IHS Climate Finance	-0.000	0.004	-0.006
	0.004	0.003	0.004
adj. R^2	0.068	0.239	0.068
Controls	✓	✓	✓
FE	✓	✓	✓
Observations	500.000	500.000	475.000

Climate finance is in mil. 2023 USD and capital formation in 2023 USD. No scale factor was applied variables that were IHS transformed. The L indicates a lag of one year. All regressions include country and year fixed effects. Standard errors clustered at the country level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

[2] I log Public and Private Capital Formation since there are no zeros present in the data, and I opt for an IHS transformation because there are zeros present.

In Table 5.16, I estimate the interaction between climate finance and ND-GAIN. I estimate specifications for the impact for climate finance commitments in the same year (1), climate finance lagged one year (2), and accumulated climate finance (3). The marginal effect of climate finance dependent on ND-GAIN is statistically significant and negative for each, with the largest coefficient observed in specification (3). This may indicate that climate finance influences capital accumulation more strongly in countries that are more vulnerable and have not adapted to climate change.

Table 5.16: Climate Finance, Adaptation Readiness (ND-GAIN), and Gross Capital Formation

	(1)	(2)	(3)
Climate Finance (IHS)	0.095***		
	0.023		
ND-GAIN	0.056***		0.191***
	0.018		0.052
Climate Finance (IHS) × ND-GAIN	-0.002***		
	0.001		
L.ND-GAIN		0.029	
		0.019	
L.Climate Finance (IHS)		0.058***	
		0.019	
L.Climate Finance (IHS) × L.ND-GAIN		-0.001***	
		0.000	
Cumulative Climate Finance (IHS)			0.410***
			0.099
Cumulative Climate Finance (IHS) × ND-GAIN			-0.008***
			0.002
adj. R^2	0.668	0.653	0.693
Controls	✓	✓	✓
FE	✓	✓	✓
Observations	586.000	586.000	586.000

Climate finance is in mil. 2021 USD and is IHS transformed. All regressions include country and year fixed effects (FE). Standard errors clustered at the country level. *, **, *** denote significance at the 10, 5, and 1 percent levels.

5.0.7 Discussion

Three key themes permeate these results.

First, the associated increase in gross capital formation in the subsequent year and long run (using accumulated climate finance) supports the idea that climate finance can improve capital formation. The sizable difference in effect size in specifications utilizing cumulative climate finance and a one-year lag highlights the importance of long-term investment in improving investor expectations. The following heterogeneity analysis provides evidence that the capital accumulation is largely driven by the private sector, and it also underscores the lack of any sizable impact resulting from the potential de-risking of investment upon the introduction of climate finance as hypothesized by Stern and Stiglitz (2023).

Second, across all regressions, there are nuanced labor market impacts as a result of climate finance. At an aggregate level, in the near term, we may observe frictional unemployment as we observe increased employment in the near term. Across the economy, I also find a large relative discrepancy between labor force participation between prime age males and prime age women. This is indicative of the types of employment from climate finance that possibly benefits men at the cost of crowding out women (Zhao et al., 2025). Climate finance does not address gender-related social norms, and any improvement in household income from male employment may lead women to leave the labor force rather than take low-quality jobs (Gasparini & Marchionni, 2017; Zhao et al., 2025). More experienced workers may benefit from these changes by filling in gaps in the labor market.

Third, the regressions from interactions between climate finance and ND-GAIN are also indicative that the impact of climate finance on capital accumulation and unemployment may be more potent for economies that are more vulnerable to climate change. Estimated effects on infant mortality also potentially indicate that any additional benefits gained from this vulnerability are not gained by addressing the health impacts of climate change. Instead, I suggest that financing constraints increase the marginal effect of climate finance compared to established markets (Ameli et al., 2021).

However, it is important to note when interpreting these results the limitations of TWFE as a model, notably dynamic panel bias (Klosin, 2024; Nickell, 1981). The introduction of fixed effects can lead to biased and inconsistent estimates because the lagged outcome is now correlated to the error term, creating endogeneity in the model. Therefore, in the following discussion, I note that the results should be interpreted as correlations rather than casually. Additionally, the interaction term testing with ND-

GAIN shows there may be treatment heterogeneity; when using TWFE, this may produce misleading estimates for a variety of reasons (de Chaisemartin & D'Haultfœuille, 2023). Lastly, there may be endogeneity stemming from omitted variable bias, and TWFE may fail to capture long-run relationships because it pools across all time differences (Ishimaru, 2025). Addressing these limitations would require using alternative methods such as Staggered DiD, IV, or GMM.

The aforementioned results and note about limitations can explain subsequent influence on the output and poverty. The modest gains in capital formation and changes in the labor market may contribute to a decrease in the output per capita growth rate. Given the associated effects on capital formation and employment, it is reasonable to expect the lack of association with poverty statistics.

Chapter 6

Conclusion

This thesis estimates the impact of climate finance on key poverty vectors. Relative to other articles on climate finance, it utilizes a novel dataset and tests impacts collectively on a number of human development measures. The results highlight the importance of the modest impact of the private sector in enacting climate finance investments. Additionally, while climate finance may induce some employment, these outcomes are heterogeneous across subgroups of workers, with the highest cost born by prime age women. Lastly, given that vulnerability to climate change is a key moderating factor for these impacts, climate finance donors should increased their allocation to vulnerable countries to improve human development outcomes. However, donors should enact guidelines to equitably disburse any benefits.

Limitations with data and methodology mean these results should be interpreted with caution. Labor force participation rate estimations samples were small and restricted only to OECD countries given data availability. Additionally, alternative estimation techniques such as Staggered DiD, IV, or GMM could be applied to address potential endogeneity, which should be explored in future research. Furthermore, future research should continue to investigate the specific elements of climate change vulnerability that may contribute to the results.

Appendix

Appendix A

Alternative Specifications

Table A.1: Climate Finance (CF), ND-GAIN, and Output with Lags

	(1)	(2)	(3)	(4)
IHS CF	1.511			
	1.452			
ND-GAIN	0.645			
	0.739			
IHS CF × ND-GAIN	-0.032			
	0.031			
L.ND-GAIN		-0.085		
		0.444		
L.IHS CF		0.116		
		0.763		
L5.IHS CF			0.637	
			0.832	
L10.IHS CF				-0.261
				0.453
L.IHS CF × L.ND-GAIN		-0.007		
		0.016		
L5.ND-GAIN			0.320	
			0.488	
L5.IHS CF × L5.ND-GAIN			-0.015	
			0.018	
L10.ND-GAIN				0.450
				0.388
L10.IHS CF × L10.ND-GAIN				0.006
				0.010
Controls	✓	✓	✓	✓
FE	✓	✓	✓	✓
Observations	357	357	318	236

Table A.2: Climate Finance per Capita (CFPC), Adaptation Readiness (ND-GAIN), and Infant Mortality

	(1)	(2)	(3)	(4)	(5)
IHS CFPC	0.001				
	0.004				
ND-GAIN	-0.000	-0.001			
	0.001	0.001			
IHS CFPC × ND-GAIN	-0.000				
	0.000				
IHS Cumulative CFPC × ND-GAIN		0.000			
		0.000			
L.ND-GAIN			-0.000		
			0.001		
L.IHS CFPC			0.003		
			0.003		
L5.IHS CFPC				-0.003	
				0.006	
L10.IHS CFPC					0.018
					0.012
L.IHS CFPC × L.ND-GAIN			-0.000		
			0.000		
L5.ND-GAIN				-0.000	
				0.001	
L5.IHS CFPC × L5.ND-GAIN				0.000	
				0.000	
L10.ND-GAIN					0.000
					0.001
L10.IHS CFPC × L10.ND-GAIN					-0.000
					0.000
Controls	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓
Overall R^2	0.993	0.993	0.993	0.993	0.992
Observations	2303	2303	2303	2081	1515

Table A.3: Short Run Effect of Climate Finance on Poverty with ND-GAIN Interaction

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable	Poverty Rate		Poverty Gap		Squared Gap	
Cumulative Climate Finance (IHS)	0.004	-0.014***	0.001	-0.008***	0.001	-0.006***
	0.008	0.004	0.005	0.002	0.003	0.002
ND-GAIN	0.006	-0.013***	0.002	-0.008***	0.002	-0.006***
	0.008	0.004	0.004	0.002	0.003	0.001
Cumulative Climate Finance (IHS) × ND-GAIN	-0.000	0.000***	-0.000	0.000***	-0.000	0.000***
	0.000	0.000	0.000	0.000	0.000	0.000
Controls	✓	✓	✓	✓	✓	✓
FE		✓		✓		✓
Observations	163	163	163	163	163	163

Table A.4: Climate Finance on Poverty with ND-GAIN Interaction and One-Year Lag

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable	Poverty Rate		Poverty Gap		Squared Gap	
L.Climate Financing (IHS)	0.017*	-0.002	0.004	-0.004*	0.001	-0.004**
	0.009	0.004	0.003	0.002	0.002	0.002
L.ND-GAIN	0.012**	-0.005*	0.004**	-0.004**	0.002	-0.003**
	0.004	0.002	0.001	0.001	0.001	0.001
L.Climate Financing (IHS) × L.ND-GAIN	-0.000*	0.000	-0.000	0.000*	-0.000	0.000**
	0.000	0.000	0.000	0.000	0.000	0.000
Controls	✓	✓	✓	✓	✓	✓
FE		✓		✓		✓
Observations	130	130	130	130	130	130

Table A.5: Long Run Effect of Climate Finance on Poverty with ND-GAIN Interaction

	(1)	(2)	(3)	(4)	(5)	(6)
Dependent Variable	Poverty Rate		Poverty Gap		Squared Gap	
Cumulative Climate Finance (IHS)	0.004	-0.014***	0.001	-0.008***	0.001	-0.006***
	0.008	0.004	0.005	0.002	0.003	0.002
ND-GAIN	0.006	-0.013***	0.002	-0.008***	0.002	-0.006***
	0.008	0.004	0.004	0.002	0.003	0.001
Cumulative Climate Finance (IHS) × ND-GAIN	-0.000	0.000***	-0.000	0.000***	-0.000	0.000***
	0.000	0.000	0.000	0.000	0.000	0.000
Controls	✓	✓	✓	✓	✓	✓
FE		✓		✓		✓
Observations	163	163	163	163	163	163

Appendix B

Relevant Computer Output

```
-------------------------------------------------------------------------------
      name:  <Ethan Hunt>
       log:  C:/Users/antho/OneDrive/Climate Finance/Code/all_regressions.txt
  log type:  text
 opened on:   4 Mar 2026, 18:04:01

.
.
.
. local files : dir "‘basedir’" files "*.do"


.
.
.
. foreach f of local files {
  2.
.
.
.     cd "‘basedir’"
  3.
```

```
.
.
.     di "--------------------------------------------------"
  4.
.     di "Running file: ‘f’"
  5.
.     di "--------------------------------------------------"
  6.
.
.
.     do "‘basedir’/‘f’"
  7.
. }
C:\Users\antho\OneDrive\Climate Finance\Code
--------------------------------------------------
Running file: 2.1.1 output lag.do
--------------------------------------------------


. *Ethan Hunt
. *Entity Effects Regression - Capital Formation
.
. *Drawing heavily from Hasen and Tarp 2001 w/ this estimation
.
. set more off

. cd "C:\Users\antho\OneDrive\Climate Finance"
C:\Users\antho\OneDrive\Climate Finance

. use "Output\Data\cf_data_v4.dta", clear

.
. encode code, gen(code_num)
```

```
. xtset code_num yr

Panel variable: code_num (strongly balanced)
 Time variable: yr, 2000 to 2023
         Delta: 1 unit

.
. *GDP Variables
. sort code

. gen gdppc = gdp/population
(109 missing values generated)

. by code: gen i_gdppc = gdppc[1]
(96 missing values generated)

.
. sort code_num yr

. by code_num: gen gdppc_g = 100 * (gdppc - L.gdppc) / L.gdppc
(291 missing values generated)

.
. *IHS Transforming Climate Finance (levels) (real 2023 USD)
. sort country yr

. bysort country (yr): gen cum_cf = sum(Financing)

. gen ihs_cum_cf = asinh(cum_cf)

. gen ihs_cf = asinh(Financing)
```

```
.
. *Working variable
. *Needed to keep esttab output on the same line.
.
. sort code_num yr

. eststo a0: reg gdppc_g ihs_cf, vce(cluster code_num)

Linear regression                               Number of obs     =      4,149
                                                F(1, 181)         =      20.93
                                                Prob > F          =     0.0000
                                                R-squared         =     0.0061
                                                Root MSE          =     13.382

                             (Std. err. adjusted for 182 clusters in code_num)
------------------------------------------------------------------------------
             |               Robust
     gdppc_g | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      ihs_cf |  -.1211062    .026474    -4.57   0.000    -.1733435   -.0688689
       _cons |   5.293138   .4417565    11.98   0.000     4.421483    6.164793
------------------------------------------------------------------------------

. estadd local ctrl     "": a0

. estadd local fe      "": a0

.
. label variable ihs_cf "IHS Climate Finance"

. label variable ihs_cum_cf "IHS Cumulative Climate Finance"
```

```
.
. eststo a1: reg gdppc_g ihs_cf aid inflation l.gdppc_g pol private_flows cap_fo
> rm sch budget m2, vce(cluster code_num)

Linear regression                               Number of obs     =        357
                                                F(9, 36)          =          .
                                                Prob > F          =          .
                                                R-squared         =     0.1674
                                                Root MSE          =     9.6669

                               (Std. err. adjusted for 37 clusters in code_num)
-------------------------------------------------------------------------------
              |               Robust
      gdppc_g | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
--------------+----------------------------------------------------------------
       ihs_cf |  -.0737029   .1430796    -0.52   0.610    -.3638817     .216476
          aid |  -4.11e-10   6.29e-10    -0.65   0.518    -1.69e-09    8.65e-10
    inflation |  -.0860265   .1368341    -0.63   0.534    -.3635388    .1914859
              |
      gdppc_g |
          L1. |   .1818296   .0509763     3.57   0.001      .078445    .2852143
              |
          pol |   .3453034   .7952551     0.43   0.667    -1.267549    1.958155
private_flows |   .0000317   .0000519     0.61   0.546    -.0000737     .000137
     cap_form |   5.54e-13   5.87e-13     0.94   0.352    -6.37e-13    1.74e-12
          sch |  -.8615124   .3542277    -2.43   0.020     -1.57992   -.1431053
       budget |   1.135378    .235842     4.81   0.000     .6570678    1.613687
           m2 |   .0061945   .0247845     0.25   0.804    -.0440708    .0564597
        _cons |   14.74602   3.913024     3.77   0.001     6.810036      22.682
-------------------------------------------------------------------------------
```

```
. estadd local ctrl     "\ding{51}": a1

. estadd local fe      "": a1

.
. *Had to take out initial gdp because of country fixed effects
. eststo a2: xtreg gdppc_g ihs_cf aid inflation l.gdppc_g pol private_flows cap_
> form sch budget m2 i.yr, fe vce(cluster code_num)

Fixed-effects (within) regression               Number of obs     =        357
Group variable: code_num                        Number of groups  =         37

R-squared:                                      Obs per group:
     Within  = 0.5550                                         min =          1
     Between = 0.0069                                         avg =        9.6
     Overall = 0.2981                                         max =         22

                                                F(29, 36)         =          .
corr(u_i, Xb) = -0.6048                         Prob > F          =          .

                               (Std. err. adjusted for 37 clusters in code_num)
------------------------------------------------------------------------------
             |               Robust
     gdppc_g | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      ihs_cf |   .0164199   .1370541     0.12   0.905    -.2615386    .2943784
         aid |  -2.19e-09   8.00e-10    -2.73   0.010    -3.81e-09   -5.63e-10
   inflation |  -.3386698   .1539376    -2.20   0.034    -.6508698   -.0264698
             |
     gdppc_g |
         L1. |   .0598049   .0883068     0.68   0.503    -.1192896    .2388994
             |
```

```
          pol |  -.198436   1.249258    -0.16   0.875    -2.732049    2.335177
private_flows |  .0000449   .0000587     0.76   0.449    -.0000741    .0001638
     cap_form |  3.33e-12   1.39e-12     2.40   0.022     5.21e-13    6.15e-12
          sch |  1.295509   1.208008     1.07   0.291    -1.154445    3.745463
       budget |  .6152889   .4147654     1.48   0.147    -.2258944    1.456472
           m2 | -.2604998   .0728589    -3.58   0.001    -.4082645   -.1127351
              |
           yr |
         2003 |   6.64023   5.112318     1.30   0.202    -3.728031    17.00849
         2004 |  14.15044    3.64369     3.88   0.000     6.760691    21.54018
         2005 |  14.56932   4.726713     3.08   0.004     4.983101    24.15554
         2006 |  10.83461   3.066633     3.53   0.001     4.615193    17.05403
         2007 |  14.58324   3.409464     4.28   0.000     7.668526    21.49795
         2008 |  14.07156   3.463259     4.06   0.000     7.047749    21.09538
         2009 | -2.575054   2.948998    -0.87   0.388    -8.555899    3.405791
         2010 |  17.25787   3.794694     4.55   0.000     9.561875    24.95387
         2011 |  12.03554   2.933518     4.10   0.000     6.086089    17.98499
         2012 |  3.832468   1.769262     2.17   0.037     .2442389    7.420697
         2013 |  7.716723   2.867462     2.69   0.011     1.901241     13.5322
         2014 |  2.027809   2.520686     0.80   0.426    -3.084379    7.139997
         2015 | -4.614647   2.364483    -1.95   0.059     -9.41004    .1807463
         2016 |  3.692161   2.751083     1.34   0.188    -1.887293    9.271615
         2017 |  7.662178    3.24825     2.36   0.024     1.074423    14.24993
         2018 |    2.6405   2.236175     1.18   0.245    -1.894673    7.175672
         2019 |  1.879653   2.533907     0.74   0.463    -3.259349    7.018655
         2020 |  -2.79041   3.278911    -0.85   0.400    -9.440351     3.85953
         2021 |  15.07655    3.75712     4.01   0.000     7.456758    22.69634
         2022 |  6.162776   3.993424     1.54   0.132    -1.936264    14.26182
         2023 |   7.61996   3.604859     2.11   0.042      .308967    14.93095
              |
        _cons |  4.256704   7.823877     0.54   0.590    -11.61085    20.12426
--------------+----------------------------------------------------------------
```

```
     sigma_u |  10.279684
     sigma_e |  7.4344762
         rho |  .65657828   (fraction of variance due to u_i)
------------------------------------------------------------------------------

. estadd local ctrl "\ding{51}": a2

. estadd local fe  "\ding{51}": a2

.
. eststo a25: xtreg gdppc_g ihs_cum_cf aid inflation l.gdppc_g pol private_flows
>  cap_form sch budget m2 i.yr, fe vce(cluster code_num)

Fixed-effects (within) regression               Number of obs     =        357
Group variable: code_num                        Number of groups  =         37

R-squared:                                      Obs per group:
     Within  = 0.5561                                         min =          1
     Between = 0.0024                                         avg =        9.6
     Overall = 0.2929                                         max =         22

                                                F(29, 36)         =          .
corr(u_i, Xb) = -0.6148                         Prob > F          =          .

                               (Std. err. adjusted for 37 clusters in code_num)
------------------------------------------------------------------------------
             |               Robust
     gdppc_g | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
  ihs_cum_cf |  -.4299975   .2956211    -1.45   0.154    -1.029545      .16955
         aid |  -2.21e-09   7.99e-10    -2.77   0.009    -3.83e-09   -5.93e-10
   inflation |  -.3308853   .1507422    -2.20   0.035    -.6366047   -.0251659
```

```
             |
     gdppc_g |
         L1. |   .0581425   .0857241     0.68   0.502     -.115714    .2319991
             |
         pol |  -.1639416   1.244019    -0.13   0.896    -2.686929    2.359046
private_flows |   .0000362    .000057     0.63   0.529    -.0000795    .0001519
    cap_form |   3.12e-12   1.43e-12     2.18   0.035     2.24e-13    6.01e-12
         sch |   1.331792   1.226107     1.09   0.285    -1.154869    3.818452
      budget |   .5892303   .4059905     1.45   0.155    -.2341565    1.412617
          m2 |  -.2637736   .0711054    -3.71   0.001    -.4079819   -.1195652
             |
          yr |
        2003 |    6.29264   5.490894     1.15   0.259     -4.84341    17.42869
        2004 |   14.34206   3.598347     3.99   0.000     7.044274    21.63985
        2005 |    15.0381   4.672939     3.22   0.003     5.560941    24.51526
        2006 |   11.43506   2.994445     3.82   0.001     5.362041    17.50807
        2007 |   15.56482   3.434763     4.53   0.000     8.598794    22.53084
        2008 |   15.20833   3.505001     4.34   0.000     8.099858     22.3168
        2009 |  -1.121886   2.778074    -0.40   0.689    -6.756081    4.512309
        2010 |   18.78801   3.714706     5.06   0.000     11.25424    26.32178
        2011 |   13.76046   2.815965     4.89   0.000     8.049418     19.4715
        2012 |   5.633786    1.92671     2.92   0.006     1.726236    9.541336
        2013 |   9.596953   2.951682     3.25   0.002     3.610665    15.58324
        2014 |   3.955894   2.477935     1.60   0.119    -1.069591    8.981379
        2015 |  -2.631079   2.133553    -1.23   0.225    -6.958124    1.695966
        2016 |   5.693884   2.835368     2.01   0.052    -.0565085    11.44428
        2017 |   9.742687    3.25056     3.00   0.005     3.150246    16.33513
        2018 |     4.7699   2.257294     2.11   0.042     .1918956    9.347904
        2019 |   4.007507   2.455369     1.63   0.111     -.972212    8.987226
        2020 |  -.6029266   2.976746    -0.20   0.841    -6.640047    5.434194
        2021 |   17.33176   3.606203     4.81   0.000     10.01804    24.64548
        2022 |   8.484198   3.795144     2.24   0.032     .7872889    16.18111
```

```
        2023 |   10.08222   3.592298     2.81   0.008     2.796705    17.36774
             |
       _cons |   12.01744   9.707693     1.24   0.224    -7.670669    31.70556
-------------+----------------------------------------------------------------
     sigma_u |  10.369022
     sigma_e |  7.4253415
         rho |  .66102115   (fraction of variance due to u_i)
------------------------------------------------------------------------------


. estadd local ctrl "\ding{51}": a25


. estadd local fe  "\ding{51}": a25


.
. eststo a3: xtreg gdppc_g l.ihs_cf aid inflation l.gdppc_g pol private_flows ca
> p_form sch budget m2 i.yr, fe vce(cluster code_num)

Fixed-effects (within) regression               Number of obs     =        357
Group variable: code_num                        Number of groups  =         37

R-squared:                                      Obs per group:
     Within  = 0.5610                                         min =          1
     Between = 0.0020                                         avg =        9.6
     Overall = 0.2899                                         max =         22

                                                F(29, 36)         =          .
corr(u_i, Xb) = -0.6215                         Prob > F          =          .

                                (Std. err. adjusted for 37 clusters in code_num)
------------------------------------------------------------------------------
             |               Robust
     gdppc_g | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
```

```
-------------+----------------------------------------------------------------
      ihs_cf |
         L1. |  -.2236057   .1245814    -1.79   0.081    -.4762685    .0290571
             |
         aid |  -2.22e-09   8.16e-10    -2.72   0.010    -3.88e-09   -5.65e-10
   inflation |  -.3147343    .139695    -2.25   0.030     -.598049   -.0314197
             |
    gdppc_g |
         L1. |   .0608069   .0835908     0.73   0.472    -.1087232    .2303369
             |
         pol |   -.013555   1.224753    -0.01   0.991    -2.497469    2.470359
private_flows |   .0000237   .0000556     0.43   0.672     -.000089    .0001365
    cap_form |   3.11e-12   1.38e-12     2.25   0.030     3.12e-13    5.91e-12
         sch |   1.234792   1.185535     1.04   0.305    -1.169584    3.639169
      budget |   .5669101   .3839018     1.48   0.148    -.2116787    1.345499
          m2 |  -.2736154   .0737782    -3.71   0.001    -.4232445   -.1239863
             |
          yr |
        2003  |   6.590599   4.986497     1.32   0.195    -3.522486    16.70368
        2004  |   14.24457   3.672328     3.88   0.000     6.796738    21.69239
        2005  |   15.47294   5.175258     2.99   0.005     4.977025    25.96885
        2006  |     11.793   3.351047     3.52   0.001     4.996761    18.58924
        2007  |   15.86666   3.950662     4.02   0.000      7.85435    23.87898
        2008  |    16.0654   4.129036     3.89   0.000     7.691328    24.43947
        2009  |  -.3321323   3.609156    -0.09   0.927     -7.65184    6.987575
        2010  |   19.74081   4.366435     4.52   0.000     10.88527    28.59635
        2011  |   14.42068   3.781303     3.81   0.001     6.751843    22.08952
        2012  |   5.872912    2.66283     2.21   0.034     .4724419    11.27338
        2013  |   10.16011   3.644669     2.79   0.008     2.768375    17.55184
        2014  |   4.487827   3.409042     1.32   0.196     -2.42603    11.40168
        2015  |   -2.16106   2.981158    -0.72   0.473    -8.207129    3.885009
        2016  |   6.110184   3.215714     1.90   0.065    -.4115856    12.63195
```

```
        2017  |   10.05371   3.810203     2.64   0.012     2.326261    17.78116
        2018  |   5.098798   3.070629     1.66   0.105    -1.128726    11.32632
        2019  |   4.585221   3.156454     1.45   0.155    -1.816364    10.98681
        2020  |  -.2655205   3.723122    -0.07   0.944    -7.816363    7.285322
        2021  |   17.85232   4.148315     4.30   0.000     9.439145    26.26549
        2022  |   9.010257   4.685883     1.92   0.062    -.4931547    18.51367
        2023  |   10.52153   4.220478     2.49   0.017        1.962    19.08105
              |
        _cons |   7.674708   7.647847     1.00   0.322    -7.835844    23.18526
--------------+----------------------------------------------------------------
      sigma_u |  10.698344
      sigma_e |  7.3839405
          rho |  .67733755   (fraction of variance due to u_i)
-------------------------------------------------------------------------------

. estadd local ctrl "\ding{51}": a3

. estadd local fe  "\ding{51}": a3

.
. eststo a4: xtreg gdppc_g l5.ihs_cf aid inflation l.gdppc_g pol private_flows c
> ap_form sch budget m2 i.yr, fe vce(cluster code_num)

Fixed-effects (within) regression               Number of obs     =        318
Group variable: code_num                        Number of groups  =         36

R-squared:                                      Obs per group:
     Within  = 0.5908                                         min =          1
     Between = 0.0025                                         avg =        8.8
     Overall = 0.2889                                         max =         19

                                                F(26, 35)         =          .
```

```
corr(u_i, Xb) = -0.6172                        Prob > F          =          .

                            (Std. err. adjusted for 36 clusters in code_num)
------------------------------------------------------------------------------
             |               Robust
    gdppc_g | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      ihs_cf |
         L5. |  -.0310841   .0885057    -0.35   0.728    -.2107603    .1485921
             |
         aid |  -1.82e-09   7.83e-10    -2.32   0.026    -3.41e-09   -2.27e-10
   inflation |  -.0453879   .3262454    -0.14   0.890    -.7077014    .6169255
             |
     gdppc_g |
         L1. |  -.0040449   .0810329    -0.05   0.960    -.1685504    .1604607
             |
         pol |   -1.05189   1.653289    -0.64   0.529    -4.408245    2.304464
private_flows |  -6.82e-06   .0000806    -0.08   0.933    -.0001705    .0001569
    cap_form |   2.85e-12   1.19e-12     2.41   0.022     4.45e-13    5.26e-12
         sch |   1.734956   1.388608     1.25   0.220    -1.084068     4.55398
      budget |   .4878829    .410897     1.19   0.243    -.3462824    1.322048
          m2 |  -.2644322     .09042    -2.92   0.006    -.4479946   -.0808698
             |
          yr |
       2006  |  -3.978986   4.436541    -0.90   0.376    -12.98564    5.027671
       2007  |   .1618233   4.376114     0.04   0.971    -8.722161    9.045807
       2008  |  -1.868019   4.197399    -0.45   0.659    -10.38919    6.653154
       2009  |  -17.29842   4.545718    -3.81   0.001    -26.52672   -8.070125
       2010  |    1.48948    5.73682     0.26   0.797    -10.15688    13.13584
       2011  |  -2.695026   4.434142    -0.61   0.547    -11.69681     6.30676
       2012  |   -10.9828   4.564988    -2.41   0.022    -20.25022   -1.715381
       2013  |  -7.905664   5.642214    -1.40   0.170    -19.35997    3.548639
```

```
        2014  |  -13.09184   4.963926    -2.64   0.012    -23.16915   -3.014535
        2015  |  -19.95947   5.639785    -3.54   0.001    -31.40884   -8.510098
        2016  |  -12.44104   5.960666    -2.09   0.044    -24.54184   -.3402469
        2017  |  -7.906498   5.902672    -1.34   0.189    -19.88956    4.076564
        2018  |   -12.5135   5.629096    -2.22   0.033    -23.94117   -1.085829
        2019  |  -13.67975     5.8083    -2.36   0.024    -25.47123   -1.888276
        2020  |  -18.49823   7.122719    -2.60   0.014    -32.95811   -4.038337
        2021  |  -1.915403   6.106948    -0.31   0.756    -14.31317    10.48236
        2022  |  -10.43391   5.012028    -2.08   0.045    -20.60887   -.2589525
        2023  |  -8.819862   6.460811    -1.37   0.181    -21.93601    4.296281
              |
       _cons  |   15.12258   10.36949     1.46   0.154    -5.928597    36.17375
--------------+----------------------------------------------------------------
     sigma_u  |  11.116701
     sigma_e  |  6.8699444
         rho  |   .7236391   (fraction of variance due to u_i)
-------------------------------------------------------------------------------

. estadd local ctrl "\ding{51}": a4

. estadd local fe  "\ding{51}": a4

.
. eststo a5: xtreg gdppc_g l10.ihs_cf aid inflation l.gdppc_g pol private_flows
> cap_form sch budget m2 i.yr, fe vce(cluster code_num)

Fixed-effects (within) regression               Number of obs     =        236
Group variable: code_num                        Number of groups  =         35

R-squared:                                      Obs per group:
     Within  = 0.6024                                         min =          1
     Between = 0.0053                                         avg =        6.7
```

```
     Overall = 0.3101                                          max =          14

                                              F(21, 34)         =           .
corr(u_i, Xb) = -0.5942                       Prob > F          =           .

                               (Std. err. adjusted for 35 clusters in code_num)
------------------------------------------------------------------------------
              |               Robust
     gdppc_g | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      ihs_cf |
        L10. |   .0148976   .0495884     0.30   0.766    -.0858781    .1156732
             |
         aid |  -1.67e-09   9.26e-10    -1.80   0.081    -3.55e-09    2.14e-10
   inflation |  -.1334057   .3036629    -0.44   0.663    -.7505229    .4837115
             |
     gdppc_g |
         L1. |   .0207151   .0677549     0.31   0.762    -.1169794    .1584096
             |
         pol |   .0564893   1.804208     0.03   0.975    -3.610102     3.72308
private_flows |   .0000738   .0000934     0.79   0.435     -.000116    .0002637
    cap_form |   4.21e-12   9.00e-13     4.67   0.000     2.38e-12    6.04e-12
         sch |   2.475151   1.560878     1.59   0.122    -.6969351    5.647238
      budget |   .5719562   .4325628     1.32   0.195    -.3071171     1.45103
          m2 |  -.2047268   .0974854    -2.10   0.043     -.402841   -.0066126
             |
          yr |
       2011  |  -4.734487   2.786944    -1.70   0.098    -10.39824    .9292658
       2012  |   -13.4216   2.964604    -4.53   0.000     -19.4464   -7.396797
       2013  |  -10.35063   2.748192    -3.77   0.001    -15.93563   -4.765636
       2014  |  -15.82376    2.83036    -5.59   0.000    -21.57575   -10.07178
       2015  |  -22.56228   3.453768    -6.53   0.000    -29.58118   -15.54337
```

```
        2016  |   -14.9757   2.851987    -5.25   0.000    -20.77164   -9.179764
        2017  |  -10.64344   2.196616    -4.85   0.000    -15.10751   -6.179383
        2018  |  -15.09996   3.273494    -4.61   0.000     -21.7525   -8.447418
        2019  |  -17.07905   3.524801    -4.85   0.000    -24.24231   -9.915792
        2020  |   -22.0828   4.147525    -5.32   0.000    -30.51159   -13.65402
        2021  |  -4.921283    4.12162    -1.19   0.241    -13.29742    3.454856
        2022  |  -14.32269   4.649891    -3.08   0.004     -23.7724   -4.872972
        2023  |  -12.47962   4.629274    -2.70   0.011    -21.88744   -3.071809
              |
       _cons  |   7.779957   13.67576     0.57   0.573    -20.01253    35.57244
--------------+----------------------------------------------------------------
     sigma_u  |  9.7705982
     sigma_e  |  6.0730154
         rho  |  .72132526   (fraction of variance due to u_i)
-------------------------------------------------------------------------------


. estadd local ctrl "\ding{51}": a5


. estadd local fe  "\ding{51}": a5


.
. esttab a0 a1 a2 a25 a3 a4 a5 using "Output\Tables\output lag.tex", replace ///
>         fragment unstack label ///
>         starlevels(* 0.10 ** 0.05 *** 0.01) ///
>     cells(b(star fmt(3)) se(fmt(3))) ///
>         keep(*ihs_cf* *ihs_cum_cf*) ///
>     booktabs ///
>         collabels(none) ///
>         nomtitles nodepvars noobs ///
>          stats(r2_a ctrl fe N, ///
>          fmt(3 %s %s 0) ///
>     labels("adj. $ R^2$ ""Controls" "FE" "Observations")) ///
```

```
>         prehead("\begin{table}[htbp]\centering \def\sym#1{\ifmmode^{#1}\else\(
> ^{#1}\)\fi} \caption{IHS Climate Finance and GDP per Capita Growth} \begin{adj
> ustbox}{max width=\textwidth}  \begin{tabular}{l*{7}{c}} \toprule") ///
> postfoot("\bottomrule \end{tabular} \end{adjustbox}  \begin{center}\begin{mini
> page}{0.98\textwidth} \footnotesize \flushleft Climate finance is in mil. 2021
>  USD and is IHS transformed. The L refers to the number of lags. FE refers to
> country and year fixed effects. Standard errors clustered at the country level
> . *, **, *** denote significance at the 10, 5, and 1 percent levels. \end{mini
> page} \end{center}\end{table}")
(output written to Output\Tables\output lag.tex)


.
. /*
>
> r2_a option note
> https://www.statalist.org/forums/forum/general-stata-discussion/general/146111
> 5-is-e-r2_a-the-adjusted-within-r-squared-in-xtreg-fe
>
> Interesting note from the stata forum
> Alvaro:
> welcome to this forum.
> -r- is the abbreviation of -robust- (ie, cluster-robust standard error).
> Interestingly, while you can invoke it via -robust- or -vce(cluster clusterid)
> - in -xtreg- (as both these options do the very same job, that is taking heter
> oskedasticity and/or autocorrelation into account), in -regress- the -robust-
> option takes heteroskedasticity only into account, whereas -.vce(cluster clust
> erid- does the same with autocorrelation of the epsilon

end of do-file

C:\Users\antho\OneDrive\Climate Finance\Code
--------------------------------------------------
```

```
Running file: 2.1.2 output lag int.do
--------------------------------------------------


. *Ethan Hunt
. *Entity Effects Regression - Capital Formation
.
. *Drawing heavily from Hasen and Tarp 2001 w/ this estimation
.
. set more off

. cd "C:\Users\antho\OneDrive\Climate Finance"
C:\Users\antho\OneDrive\Climate Finance

. use "Output\Data\cf_data_v4.dta", clear


.
. encode code, gen(code_num)

. xtset code_num yr

Panel variable: code_num (strongly balanced)
 Time variable: yr, 2000 to 2023
         Delta: 1 unit


.
. *GDP Variables
. sort code

. gen gdppc = gdp/population
(109 missing values generated)

. by code: gen i_gdppc = gdppc[1]
```

```
(96 missing values generated)

.
. sort code_num yr

. by code_num: gen gdppc_g = 100 * (gdppc - L.gdppc) / L.gdppc
(291 missing values generated)

.
. *IHS Transforming Climate Finance (levels) (real 2023 USD)
. sort country yr

. bysort country (yr): gen cum_cf = sum(Financing)

. gen ihs_cum_cf = asinh(cum_cf)

. gen ihs_cf = asinh(Financing)

.
. sort code_num yr

.
. eststo a1: xtreg gdppc_g c.ihs_cf##c.gain aid inflation l.gdppc_g pol private_
> flows cap_form sch budget m2 i.yr, fe vce(cluster code_num)

Fixed-effects (within) regression               Number of obs     =        357
Group variable: code_num                        Number of groups  =         37

R-squared:                                      Obs per group:
     Within  = 0.5577                                         min =          1
     Between = 0.0069                                         avg =        9.6
     Overall = 0.3011                                         max =         22
```

```
                                                F(29, 36)         =          .
corr(u_i, Xb) = -0.5980                         Prob > F          =          .

                              (Std. err. adjusted for 37 clusters in code_num)
------------------------------------------------------------------------------
             |               Robust
    gdppc_g | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      ihs_cf |   1.510855   1.452094     1.04   0.305    -1.434127    4.455838
        gain |   .6452472   .7385891     0.87   0.388    -.8526809    2.143175
             |
  c.ihs_cf#|
      c.gain |  -.0317973   .0310044    -1.03   0.312    -.0946772    .0310826
             |
         aid |  -2.15e-09   8.19e-10    -2.62   0.013    -3.81e-09   -4.86e-10
   inflation |  -.3167706   .1612892    -1.96   0.057    -.6438802     .010339
             |
     gdppc_g |
         L1. |   .0654386   .0898286     0.73   0.471    -.1167422    .2476193
             |
         pol |   -.760934   1.339492    -0.57   0.574    -3.477549    1.955681
private_flows |   .0000506   .0000598     0.84   0.404    -.0000708    .0001719
    cap_form |   3.56e-12   1.45e-12     2.45   0.019     6.18e-13    6.50e-12
         sch |   1.196265    1.21988     0.98   0.333    -1.277767    3.670297
      budget |   .6091811   .4080992     1.49   0.144    -.2184825    1.436845
          m2 |   -.266225   .0768745    -3.46   0.001    -.4221338   -.1103163
             |
          yr |
       2003  |   7.019594   5.288425     1.33   0.193    -3.705829    17.74502
       2004  |   13.90122   3.428579     4.05   0.000     6.947738     20.8547
       2005  |   14.08081   4.703691     2.99   0.005     4.541283    23.62034
```

```
        2006  |   10.29424   2.756409     3.73   0.001     4.703985     15.8845
        2007  |   13.98641    3.19961     4.37   0.000       7.4973    20.47552
        2008  |   13.27534   3.412775     3.89   0.000     6.353914    20.19677
        2009  |  -3.073272   2.459593    -1.25   0.220    -8.061558    1.915015
        2010  |   16.70829   3.712548     4.50   0.000     9.178893    24.23768
        2011  |   11.25464   2.934567     3.84   0.000     5.303062    17.20622
        2012  |   3.225959   2.099072     1.54   0.133    -1.031156    7.483074
        2013  |   7.198656   3.191233     2.26   0.030     .7265356    13.67078
        2014  |   1.673762     2.8586     0.59   0.562    -4.123747    7.471272
        2015  |  -5.003233   2.557268    -1.96   0.058    -10.18961    .1831459
        2016  |   3.384458    2.92406     1.16   0.255     -2.54581    9.314726
        2017  |    7.19522   3.454618     2.08   0.044     .1889298    14.20151
        2018  |   2.248468   2.642249     0.85   0.400     -3.11026    7.607197
        2019  |   1.473093    2.97312     0.50   0.623    -4.556673     7.50286
        2020  |  -3.050464   3.653854    -0.83   0.409    -10.46082    4.359895
        2021  |   14.74674   3.893649     3.79   0.001     6.850055    22.64343
        2022  |   5.711164   4.204091     1.36   0.183    -2.815128    14.23746
        2023  |    7.31466   4.097083     1.79   0.083    -.9946099    15.62393
              |
        _cons |  -24.82198   37.14418    -0.67   0.508    -100.1539     50.5099
--------------+----------------------------------------------------------------
      sigma_u |  10.274164
      sigma_e |  7.4381729
          rho |  .65611173   (fraction of variance due to u_i)
-------------------------------------------------------------------------------

. estadd local cntl "\ding{51}": a1

. estadd local fe  "\ding{51}": a1

.

. eststo a2: xtreg gdppc_g c.l.ihs_cf##c.l.gain aid inflation l.gdppc_g pol priv
```

```
> ate_flows cap_form sch budget m2 i.yr, fe vce(cluster code_num)

Fixed-effects (within) regression               Number of obs     =        357
Group variable: code_num                        Number of groups  =         37

R-squared:                                      Obs per group:
     Within  = 0.5623                                         min =          1
     Between = 0.0003                                         avg =        9.6
     Overall = 0.2767                                         max =         22

                                                F(29, 36)         =          .
corr(u_i, Xb) = -0.6354                         Prob > F          =          .

                               (Std. err. adjusted for 37 clusters in code_num)
------------------------------------------------------------------------------
             |               Robust
    gdppc_g | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      ihs_cf |
         L1. |   .1163155   .7631724     0.15   0.880     -1.43147    1.664101
             |
        gain |
         L1. |  -.0848903   .4436271    -0.19   0.849    -.9846078    .8148272
             |
   cL.ihs_cf#|
    cL.gain |  -.0072714   .0159735    -0.46   0.652    -.0396673    .0251244
             |
         aid |  -2.15e-09   8.56e-10    -2.51   0.017    -3.89e-09   -4.13e-10
   inflation |  -.3118263   .1382737    -2.26   0.030    -.5922583   -.0313942
             |
    gdppc_g |
         L1. |    .057462   .0818175     0.70   0.487    -.1084715    .2233955
```

```
             |
         pol |   .0074447   1.244036     0.01   0.995    -2.515578    2.530467
private_flows |   .0000255   .0000553     0.46   0.648    -.0000866    .0001376
    cap_form |   3.40e-12   1.39e-12     2.44   0.020     5.75e-13    6.23e-12
         sch |   1.102266   1.226679     0.90   0.375    -1.385554    3.590085
      budget |   .5766189   .3798702     1.52   0.138    -.1937935    1.347031
          m2 |  -.2821236   .0736989    -3.83   0.000    -.4315919   -.1326554
             |
          yr |
       2003  |   6.518435   4.967664     1.31   0.198    -3.556455    16.59333
       2004  |    14.3322   3.594469     3.99   0.000     7.042276    21.62212
       2005  |   15.52208   5.172284     3.00   0.005     5.032207    26.01196
       2006  |   11.79624   3.429346     3.44   0.001       4.8412    18.75127
       2007  |   16.07849   3.990415     4.03   0.000     7.985549    24.17142
       2008  |   16.39734   4.210934     3.89   0.000     7.857175    24.93751
       2009  |   .0948917   3.339614     0.03   0.977    -6.678159    6.867942
       2010  |   20.17442   4.344272     4.64   0.000     11.36383    28.98501
       2011  |   15.02697   3.778564     3.98   0.000     7.363687    22.69025
       2012  |   6.442464   2.880283     2.24   0.032     .6009787    12.28395
       2013  |   10.91841   3.785161     2.88   0.007     3.241751    18.59507
       2014  |   5.288929   3.547646     1.49   0.145     -1.90603    12.48389
       2015  |  -1.399422   3.013479    -0.46   0.645    -7.511041    4.712197
       2016  |   6.759609   3.345291     2.02   0.051    -.0249558    13.54417
       2017  |   10.80485   3.823785     2.83   0.008     3.049857    18.55985
       2018  |    5.78304   3.215736     1.80   0.081    -.7387742    12.30485
       2019  |   5.533368   3.334954     1.66   0.106    -1.230232    12.29697
       2020  |   .6333663   3.844719     0.16   0.870    -7.164085    8.430817
       2021  |   18.75516   4.086488     4.59   0.000     10.46738    27.04294
       2022  |   9.972064   4.659868     2.14   0.039     .5214139    19.42271
       2023  |   11.43875   4.511831     2.54   0.016     2.288337    20.58917
             |
       _cons |    12.6296   22.63535     0.56   0.580    -33.27701    58.53621
```

```
-------------+----------------------------------------------------------------
     sigma_u |  10.805017
     sigma_e |  7.3990292
         rho |  .68077234   (fraction of variance due to u_i)
------------------------------------------------------------------------------


. estadd local cntl "\ding{51}": a2


. estadd local fe  "\ding{51}": a2


.
. eststo a3: xtreg gdppc_g c.l5.ihs_cf##c.l5.gain aid inflation l.gdppc_g pol pr
> ivate_flows cap_form sch budget m2 i.yr, fe vce(cluster code_num)

Fixed-effects (within) regression               Number of obs     =        318
Group variable: code_num                        Number of groups  =         36

R-squared:                                      Obs per group:
     Within  = 0.5922                                         min =          1
     Between = 0.0037                                         avg =        8.8
     Overall = 0.2859                                         max =         19

                                                F(28, 35)         =          .
corr(u_i, Xb) = -0.6193                         Prob > F          =          .

                               (Std. err. adjusted for 36 clusters in code_num)
------------------------------------------------------------------------------
             |               Robust
     gdppc_g | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      ihs_cf |
         L5. |   .6366984   .8323975     0.76   0.449    -1.053158    2.326555
```

```
             |
        gain |
         L5. |   .3197052   .4875628     0.66   0.516    -.6700999     1.30951
             |
 cL5.ihs_cf#|
   cL5.gain |   -.014564   .0179033    -0.81   0.421    -.0509095    .0217816
             |
         aid |  -1.83e-09   8.13e-10    -2.25   0.031    -3.48e-09   -1.82e-10
   inflation |  -.0300637   .3278447    -0.09   0.927     -.695624    .6354965
             |
    gdppc_g |
         L1. |  -.0067392    .078595    -0.09   0.932    -.1662955     .152817
             |
         pol |  -1.466956   1.612249    -0.91   0.369    -4.739997    1.806084
private_flows |  -6.07e-06   .0000776    -0.08   0.938    -.0001636    .0001514
    cap_form |   2.83e-12   1.16e-12     2.44   0.020     4.72e-13    5.20e-12
         sch |   1.682856   1.368403     1.23   0.227    -1.095149    4.460861
      budget |   .4965255   .4049797     1.23   0.228     -.325627    1.318678
          m2 |  -.2708417   .0898005    -3.02   0.005    -.4531463    -.088537
             |
          yr |
       2006  |  -4.033352   4.397532    -0.92   0.365    -12.96082    4.894111
       2007  |  -.0098987    4.50249    -0.00   0.998    -9.150439    9.130642
       2008  |  -1.905711   4.231831    -0.45   0.655    -10.49678    6.685362
       2009  |  -17.23548   4.532268    -3.80   0.001    -26.43647   -8.034486
       2010  |     1.0736   5.990693     0.18   0.859    -11.08815    13.23535
       2011  |  -2.942941   4.619737    -0.64   0.528    -12.32151    6.435624
       2012  |  -11.16174   4.659763    -2.40   0.022    -20.62156   -1.701914
       2013  |  -8.039902   5.769102    -1.39   0.172     -19.7518    3.671998
       2014  |  -13.17629   5.013879    -2.63   0.013      -23.355    -2.99757
       2015  |  -20.07361   5.707114    -3.52   0.001    -31.65967   -8.487552
       2016  |   -12.6238   6.181527    -2.04   0.049    -25.17297   -.0746363
```

```
        2017  |  -7.975634   6.070627    -1.31   0.197    -20.29966    4.348394
        2018  |  -12.63272   5.746406    -2.20   0.035    -24.29854   -.9668943
        2019  |  -13.76599    5.99035    -2.30   0.028    -25.92705   -1.604936
        2020  |  -18.50486   7.150632    -2.59   0.014    -33.02141   -3.988304
        2021  |  -2.153559   6.213428    -0.35   0.731    -14.76749    10.46037
        2022  |  -10.59889   4.978896    -2.13   0.040    -20.70659   -.4911941
        2023  |  -8.875283   6.625745    -1.34   0.189    -22.32626    4.575695
              |
        _cons |   1.218945   22.49104     0.05   0.957     -44.4403    46.87819
--------------+----------------------------------------------------------------
      sigma_u |  11.301244
      sigma_e |  6.8853809
          rho |  .72929026   (fraction of variance due to u_i)
-------------------------------------------------------------------------------

. estadd local cntl "\ding{51}": a3

. estadd local fe  "\ding{51}": a3

.
. eststo a4: xtreg gdppc_g c.l10.ihs_cf##c.l10.gain aid inflation l.gdppc_g pol
> private_flows cap_form sch budget m2 i.yr, fe vce(cluster code_num)

Fixed-effects (within) regression               Number of obs     =        236
Group variable: code_num                        Number of groups  =         35

R-squared:                                      Obs per group:
     Within  = 0.6074                                         min =          1
     Between = 0.0059                                         avg =        6.7
     Overall = 0.2805                                         max =         14

                                                F(23, 34)         =          .
```

```
corr(u_i, Xb) = -0.6500                        Prob > F          =          .

                              (Std. err. adjusted for 35 clusters in code_num)
------------------------------------------------------------------------------
             |               Robust
    gdppc_g | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     ihs_cf |
       L10. |  -.2613429   .4531454    -0.58   0.568    -1.182245    .6595594
            |
       gain |
       L10. |   .4500992   .3877734     1.16   0.254    -.3379511    1.238149
            |
 cL10.ihs_cf#|
  cL10.gain |    .006096   .0101659     0.60   0.553    -.0145637    .0267556
            |
        aid |  -1.63e-09   9.30e-10    -1.76   0.088    -3.52e-09    2.56e-10
  inflation |  -.1484215   .2851646    -0.52   0.606    -.7279458    .4311028
            |
    gdppc_g |
        L1. |   .0207859   .0650719     0.32   0.751    -.1114562     .153028
            |
        pol |   .0810392    1.66992     0.05   0.962    -3.312647    3.474725
private_flows |   .0001055   .0000947     1.11   0.273     -.000087    .0002979
   cap_form |   3.52e-12   1.08e-12     3.26   0.003     1.32e-12    5.71e-12
        sch |   2.580936   1.519823     1.70   0.099    -.5077152    5.669587
     budget |   .5850107   .4310582     1.36   0.184     -.291005    1.461026
         m2 |  -.2102949   .1007303    -2.09   0.044    -.4150035   -.0055862
            |
         yr |
       2011  |  -4.679911   2.725912    -1.72   0.095    -10.21963     .859809
       2012  |  -13.44513   2.897141    -4.64   0.000    -19.33283   -7.557436
```

```
        2013  |  -10.76507   2.630171    -4.09   0.000    -16.11022   -5.419921
        2014  |  -16.22221   2.680745    -6.05   0.000    -21.67014   -10.77428
        2015  |  -23.11181   3.304941    -6.99   0.000    -29.82826   -16.39537
        2016  |  -15.77699   2.660739    -5.93   0.000    -21.18426   -10.36972
        2017  |  -11.78761   2.099871    -5.61   0.000    -16.05506   -7.520162
        2018  |  -16.50838    3.21706    -5.13   0.000    -23.04623   -9.970527
        2019  |   -18.5582    3.42921    -5.41   0.000    -25.52719    -11.5892
        2020  |  -23.70525   3.872117    -6.12   0.000    -31.57434   -15.83616
        2021  |  -6.563628   4.077128    -1.61   0.117    -14.84935    1.722093
        2022  |  -16.19548   4.369347    -3.71   0.001    -25.07506   -7.315895
        2023  |  -14.05768   4.165404    -3.37   0.002     -22.5228   -5.592557
              |
       _cons  |  -12.33121   18.21696    -0.68   0.503    -49.35254    24.69011
--------------+----------------------------------------------------------------
     sigma_u  |  10.921177
     sigma_e  |  6.0682273
         rho  |  .76409708   (fraction of variance due to u_i)
-------------------------------------------------------------------------------


. estadd local cntl "\ding{51}": a4


. estadd local fe  "\ding{51}": a4


.
. esttab a1 a2 a3 a4 using "Output\Tables\output int.tex", replace ///
>     cells(b(star fmt(3)) se(fmt(3))) label booktabs style(tex) ///
>     keep(*ihs_cf* *gain*) ///  <-- Wildcard to keep main variables
>     coeflabel(L.ihs_cf "Lagged IHS Climate Finance (L1)" ///
>               L5.ihs_cf "Lagged IHS Climate Finance (L5)" ///
>               L10.ihs_cf "Lagged IHS Climate Finance (L10)" ///
>               ihs_cf "IHS Climate Finance" ///
>               gain "ND-GAIN Index" ///
```

```
>               L.gain "Lagged ND-GAIN Index (L1)" ///
>               L5.gain "Lagged ND-GAIN Index (L5)" ///
>               L10.gain "Lagged ND-GAIN Index (L10)" ///
>                          c.ihs_cf#c.gain "Interaction" ///
>                          c.L.ihs_cf#c.L.gain "Lagged Interaction (L1)" ///
>                          c.l5.ihs_cf#c.l5.ihs_cf "Lagged Interaction (L5)" ///
>                          c.l10.ihs_cf#c.l10.ihs_cf "Lagged Interaction (L10)")
>  ///
>     stats(cntl fe N, fmt(%s %s 0) labels("Controls" "FE" "Observations")) ///
>     nomtitles collabels(none)
(output written to Output\Tables\output int.tex)

.
. /* Keep in Results
>         xtreg gdppc_g c.l5.ihs_cum_cf##c.l5.gain aid inflation l.gdppc_g pol p
> rivate_flows cap_form sch budget m2 i.yr, fe vce(cluster code_num)

end of do-file

C:\Users\antho\OneDrive\Climate Finance\Code
--------------------------------------------------
Running file: 2.2.1 pov.do
--------------------------------------------------

. *Ethan Hunt
. *Poverty - Alvi and Senbeta
. set more off

.
. *Pov in 2021 PPP USD. CF in 2021 current USD
.
. set more off
```

```
. cd "C:\Users\antho\OneDrive\Climate Finance"

C:\Users\antho\OneDrive\Climate Finance


.

. pip, povline(2.15) clear

Version in use: 20250930_2021_01_02_PROD

variable country_name not found

r(111);


end of do-file


r(111);


.

.

.

. log close

      name:  <unnamed>

       log:  C:/Users/antho/OneDrive/Climate Finance/Code/all_regressions.txt

  log type:  text

 closed on:   4 Mar 2026, 18:04:07

--------------------------------------------------------------------------------
```

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